Digital Avionics
航電系統

Lecture 3

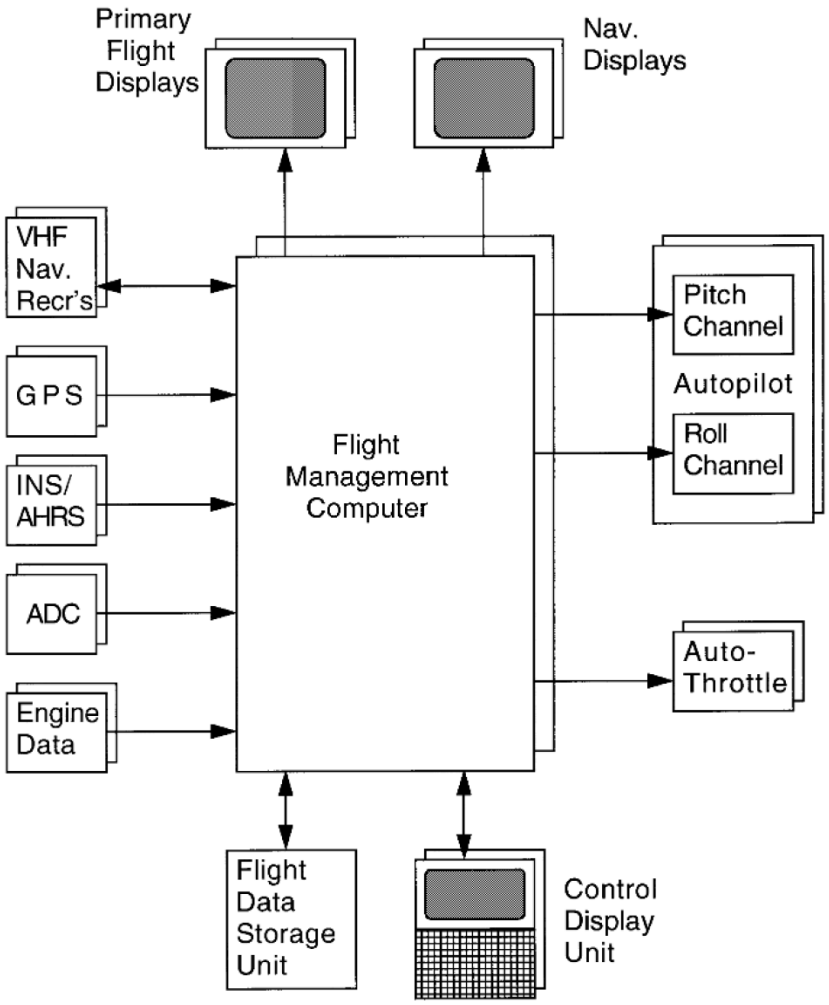
Prof. Ying-Chih Lai

Textbook

- Reading:
  - Chapter 3: Aerodynamics and Aircraft Control
  - Section 3.1~3.8, pp. 101-162

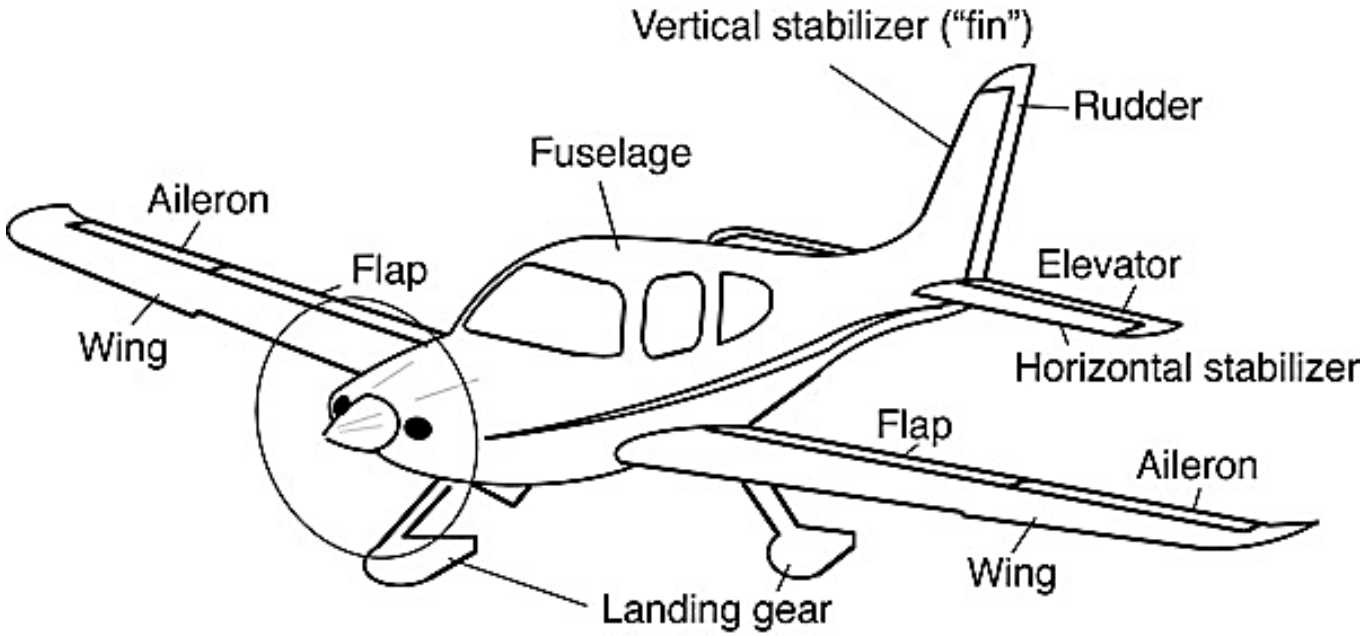
Overview of Onboard Avionics

- Flight Control System, FBW, and FMS
  - FMS is high level, A/P is middle level, and FBW is the lower level system
  - The higher level receives more external commands rather than the lower level system just carrying closed-loop performance.
  - Mini computer, microcomputer and microprocessor are used in different levels



Basic Aerodynamics

Airplane Components



Modern Control Surfaces

Rudder



Elevator



Aileron



Elevon
(elevator+aileron)

Rudderon
(rudder+aileron)



Other Devices

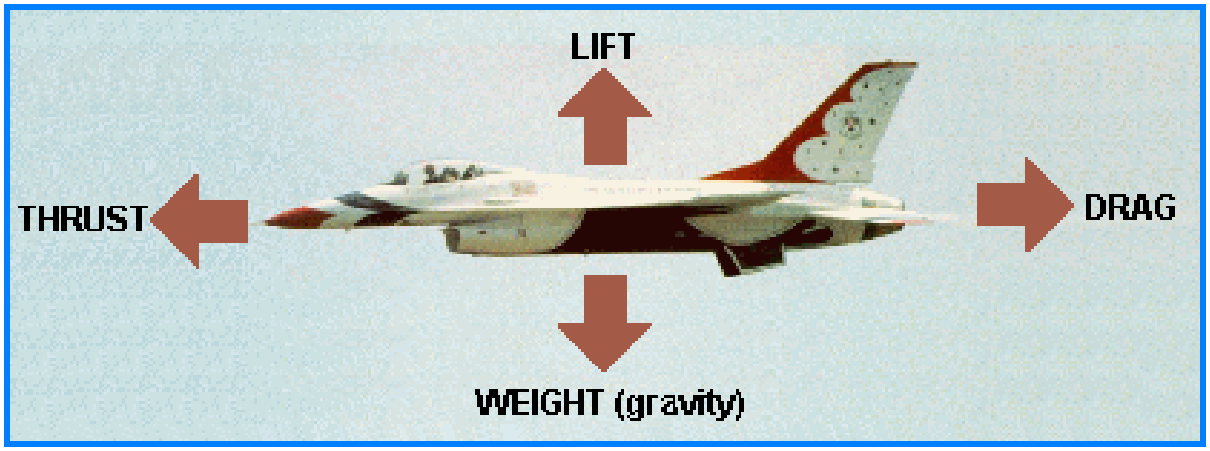




- Combinations of control surfaces and other devices: flaperons, spoilerons, decelerons (aileron and airbrake)
- Vectored thrust

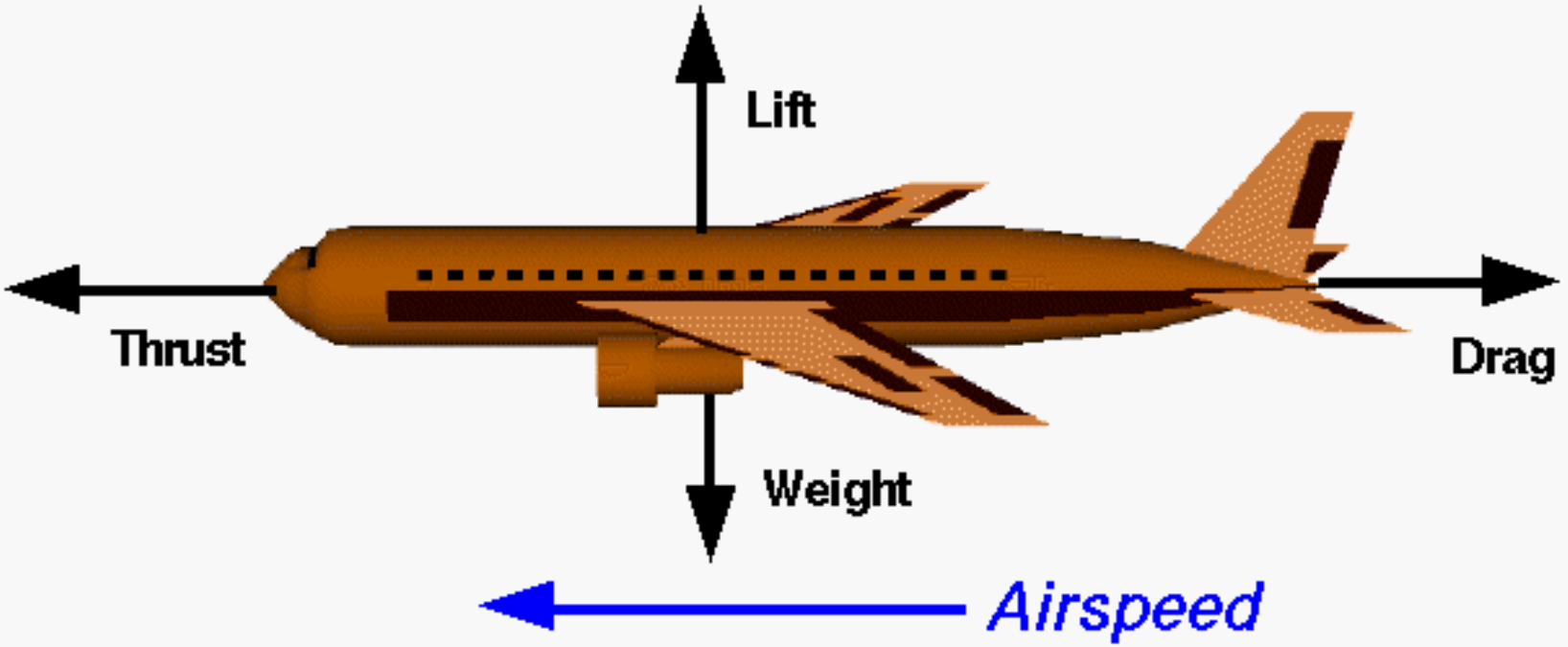
Four Forces on an Airplane

- **THRUST**: The force exerted by the engine and its propeller(s), which pushes air backward with the object of causing a reaction, or thrust, of the airplane in the forward direction.
- **DRAG**: The resistance of the airplane to forward motion directly opposed to thrust.
- **LIFT**: The upward force created by the wings moving through the air, which sustains the airplane in flight.
- **WEIGHT**: The downward force due to the weight(gravity) of the airplane and its load, directly opposed to lift.



Cruise – Balanced Forces

Glenn Research Center



Lift = Weight

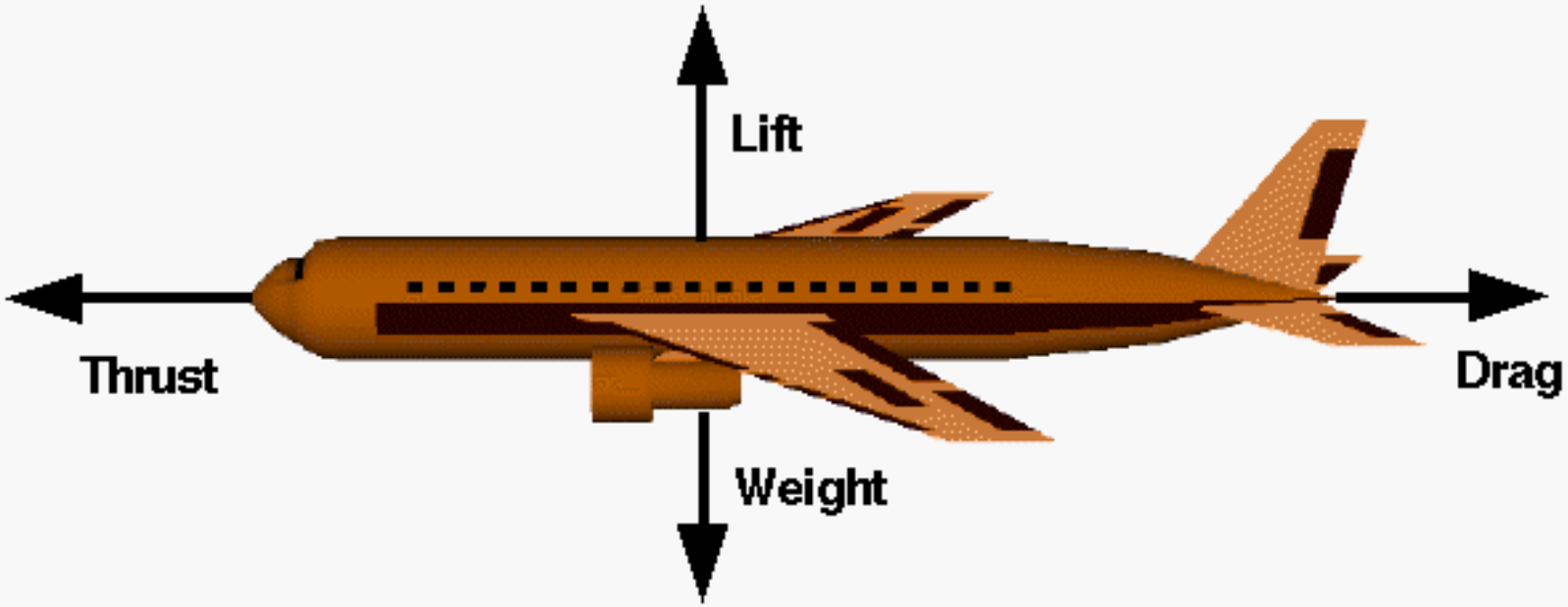
Thrust = Drag

Airplane moves in a straight line at constant airspeed.

Simplified Aircraft Motion

Unbalanced Forces

Glenn Research Center



Flight Condition	Effect
Lift > Weight	Plane Rises
Weight > Lift	Plane Falls
Drag > Thrust	Plane Slows
Thrust > Drag	Plane Accelerates

Equations of Motion

1. Lift L, which is perpendicular to the flight path direction
2. Drag D, which is parallel to the flight path direction
3. Weight W, which acts vertically toward the center of the earth (and hence is inclined at angle θ with respect to the lift direction)
4. Thrust T, which in general is inclined at the angle α_T with respect to the flight path direction

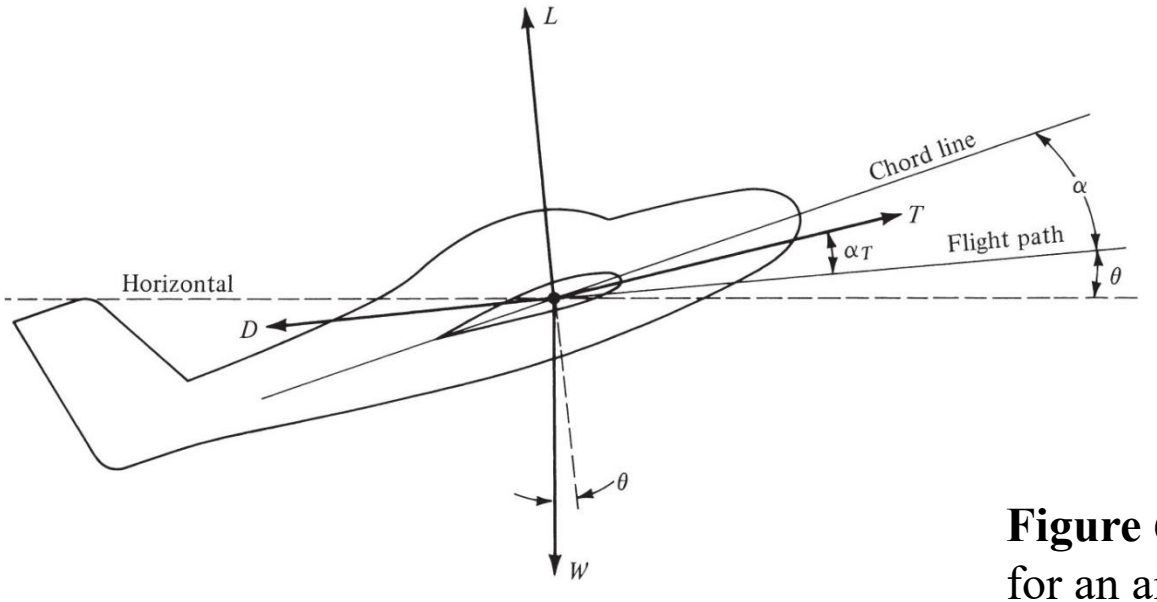
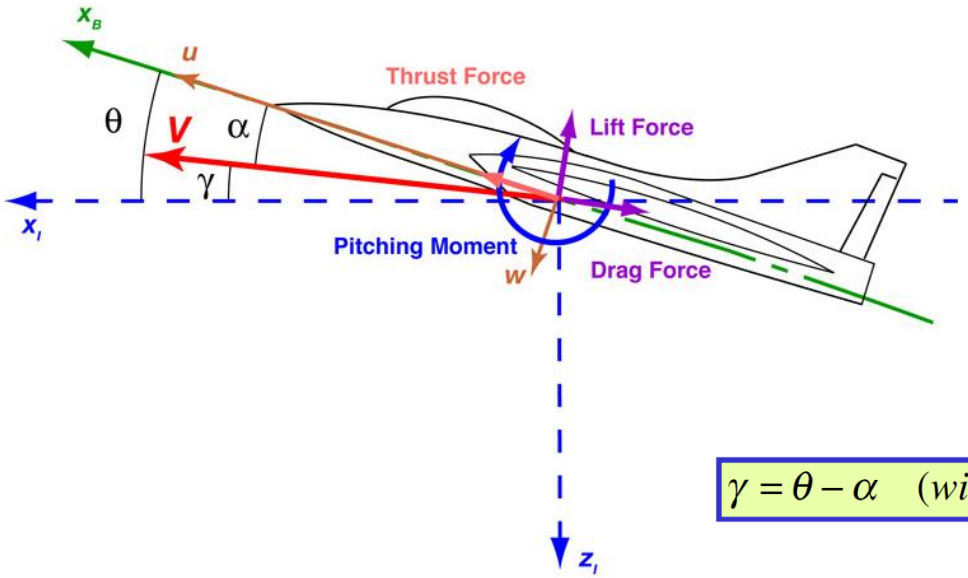


Figure 6.5 Force diagram for an airplane in flight.

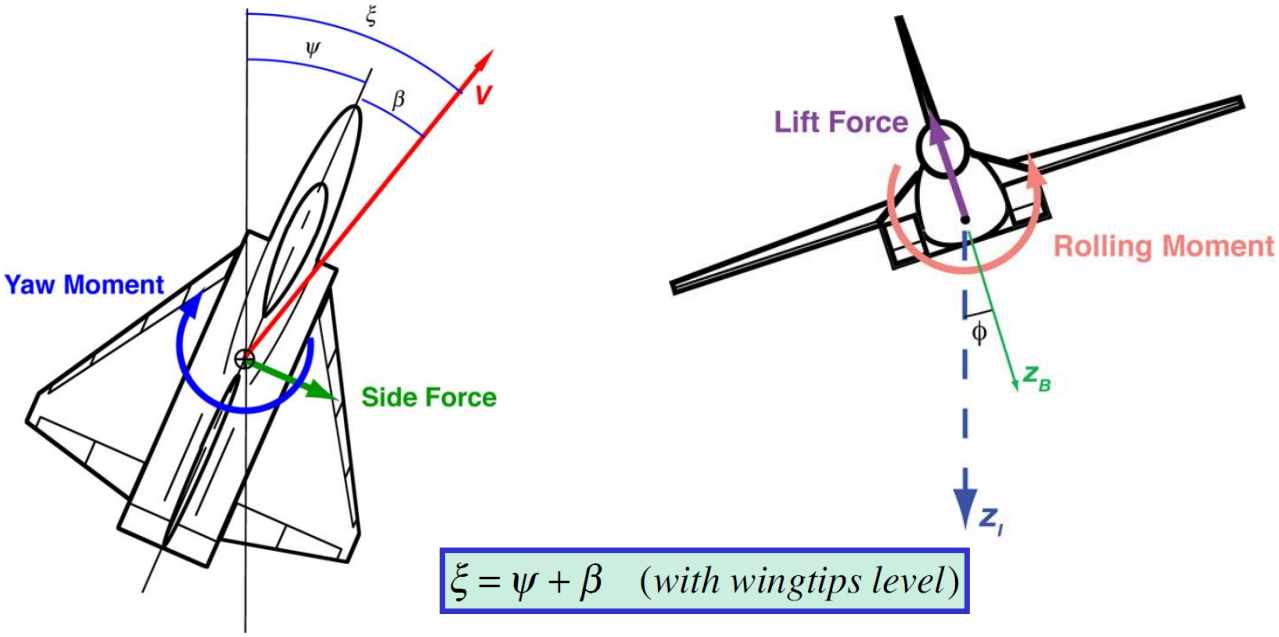
Longitudinal Variables



$\gamma = \theta - \alpha$ *(with wingtips level)*

$u(t)$: axial velocity	• along vehicle centerline
$w(t)$: normal velocity	• perpendicular to centerline
$V(t)$: velocity magnitude	• along net direction of flight
$\alpha(t)$: angle of attack	• angle between centerline and direction of flight
$\gamma(t)$: flight path angle	• angle between direction of flight and local horizontal
$\theta(t)$: pitch angle	• angle between centerline and local horizontal

Lateral-Directional Variables



$$\xi = \psi + \beta \quad (\textit{with wingtips level})$$

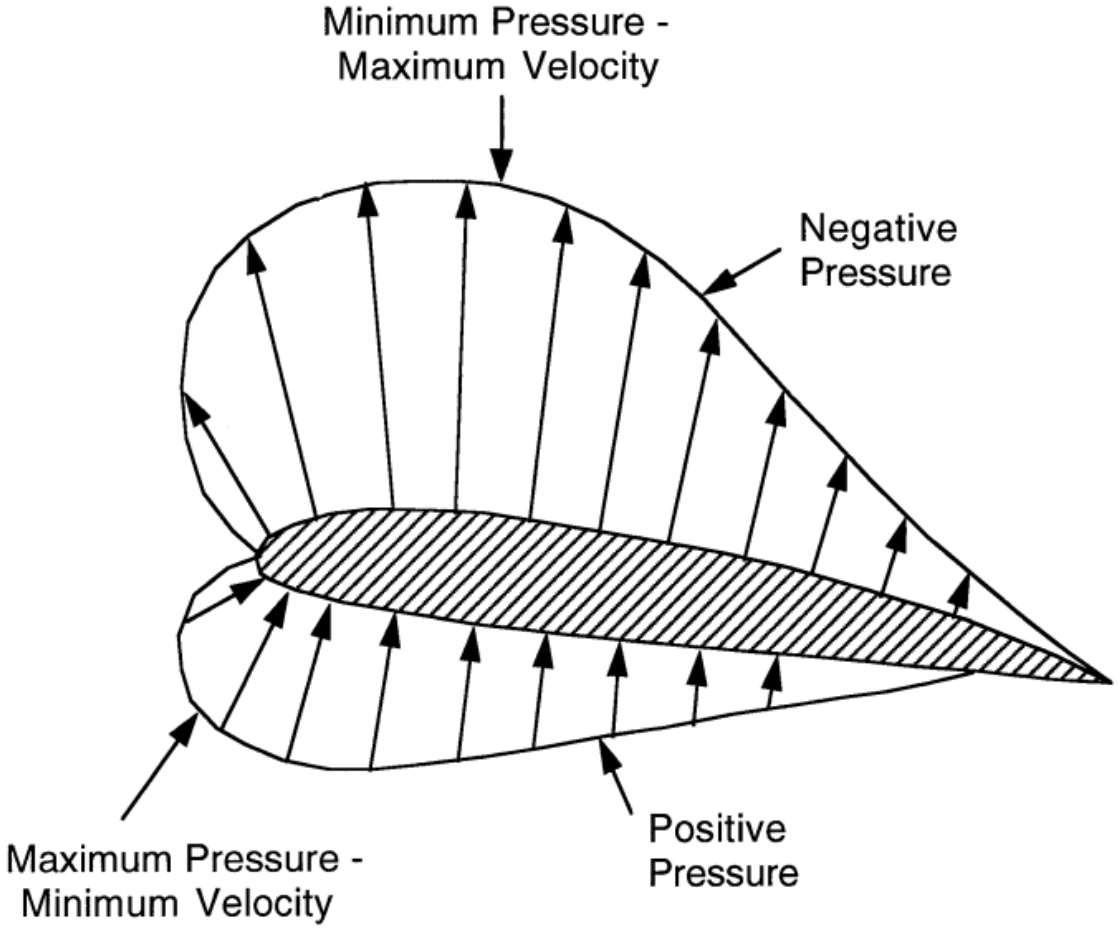
$\beta\,(t)$: sideslip angle	• angle between centerline and direction of flight
$\psi\,(t)$: yaw angle	• angle between centerline and north direction
$\xi\,(t)$: heading angle	• angle between direction of flight and compass reference (e.g., north)
$\varphi\,(t)$: roll angle	• angle between true vertical and body z axis

Aerodynamic Forces and Moments

- The physical phenomena producing lift, drag, and moments of an airplane were introduced. We emphasized that the aerodynamic forces and moments exerted on a body moving through a fluid stem from two sources:
  - 1. The pressure distribution
  - 2. The shear stress distribution
- Both acting over the body surface. The physical laws governing such phenomena were examined, with various applications to aerodynamic flows.

Pressure Distribution on Wing

- An airfoil inclined at an angle to a moving air stream will experience a resultant force due to aerodynamic effects.

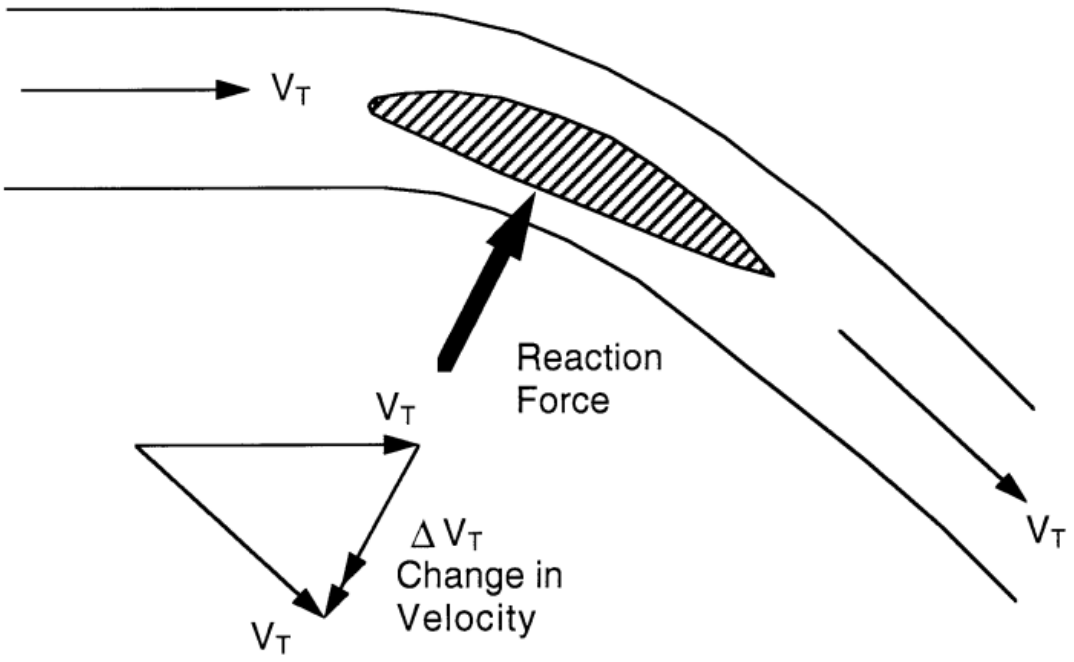


Pressure Distribution on Wing

- Air flow on the curvature wing surface to generate upper flow and lower flow in different pressure forces.
- The upper wing surface produces about 2/3 contribution to lift under maximum speed; while the lower wing surface contributes 1/3 of lift.

Interaction of Airflow to Wing

- The reaction force resulting from the rate of change of momentum of the moving air stream by the action of the airfoil in deflecting the airstream from its original direction.

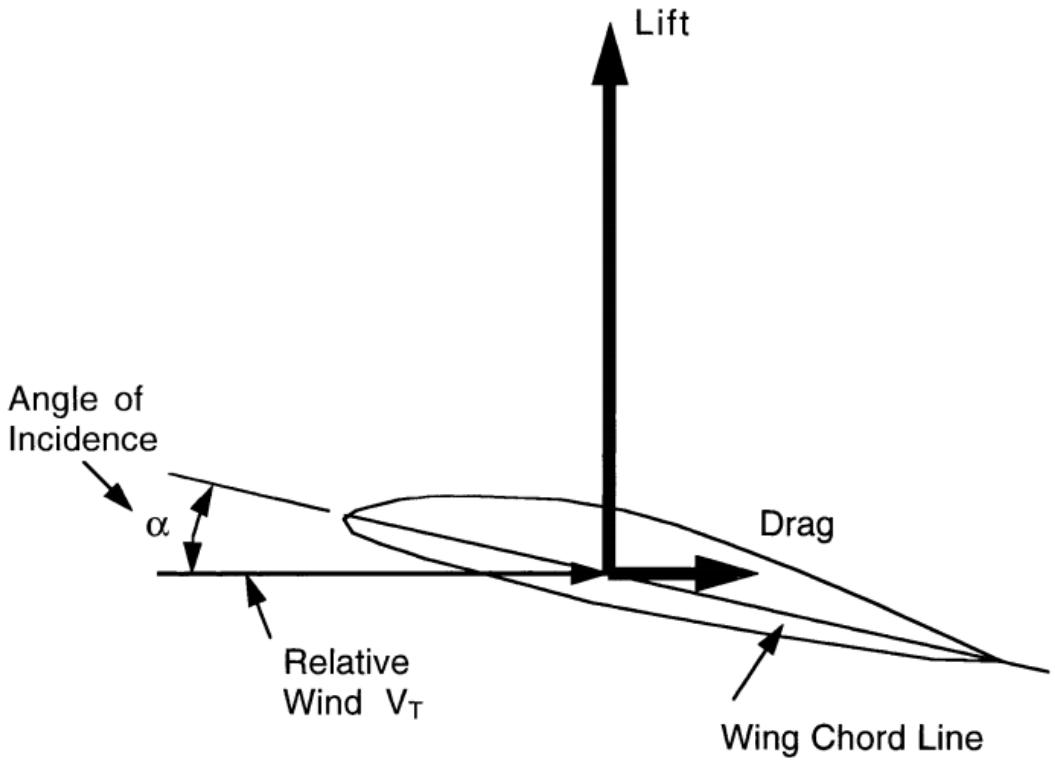


Interaction of Airflow on Wing

- Lift can be described by the airflow moment vector changes. A horizontal air flows a wing under AoA, airflow is guided to flow backward and downward.
- The downward flow is termed as down wash, and produces a combination force as ΔV_T

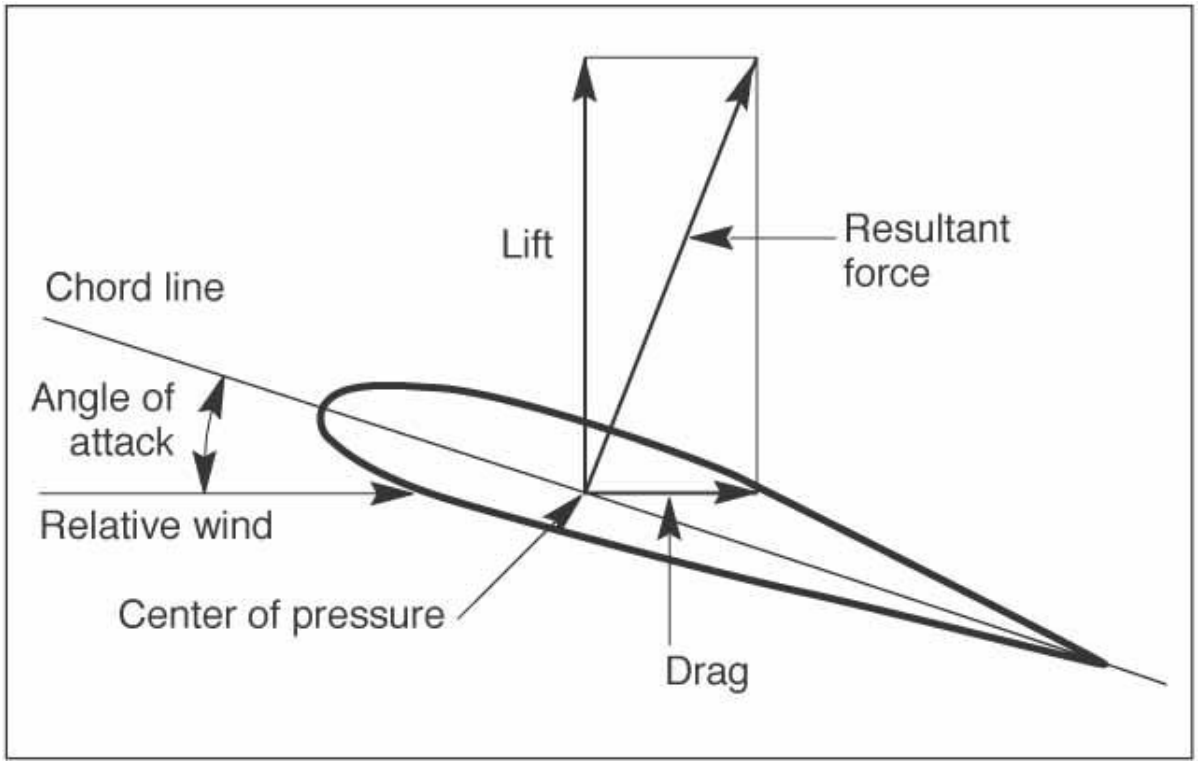
Definition of AoA

- Relative angle of airflow direction to wing chord line is defined as angle of attack (AoA) by the US system, or angle of incidence (AoI) by the British system.



Lift and Drag

- Wing Lift and Drag
  - Lift: Perpendicular to free-stream airflow
  - Drag: Parallel to the free-stream airflow (along with the inflow)



Lift Coefficient and Drag Coefficient

- **Non-dimensional force coefficients, C_L and C_D, are dimensionalized by**
  - Dynamic pressure, *q or Q, N/m2 or lb/sq ft*
  - Impact force = *QS or qS* , *S is*
  - Wing area, *S, m^2 or ft^2*

$$\text{Dynamic pressure} \quad Q = \frac{1}{2}\rho V_T{}^2$$

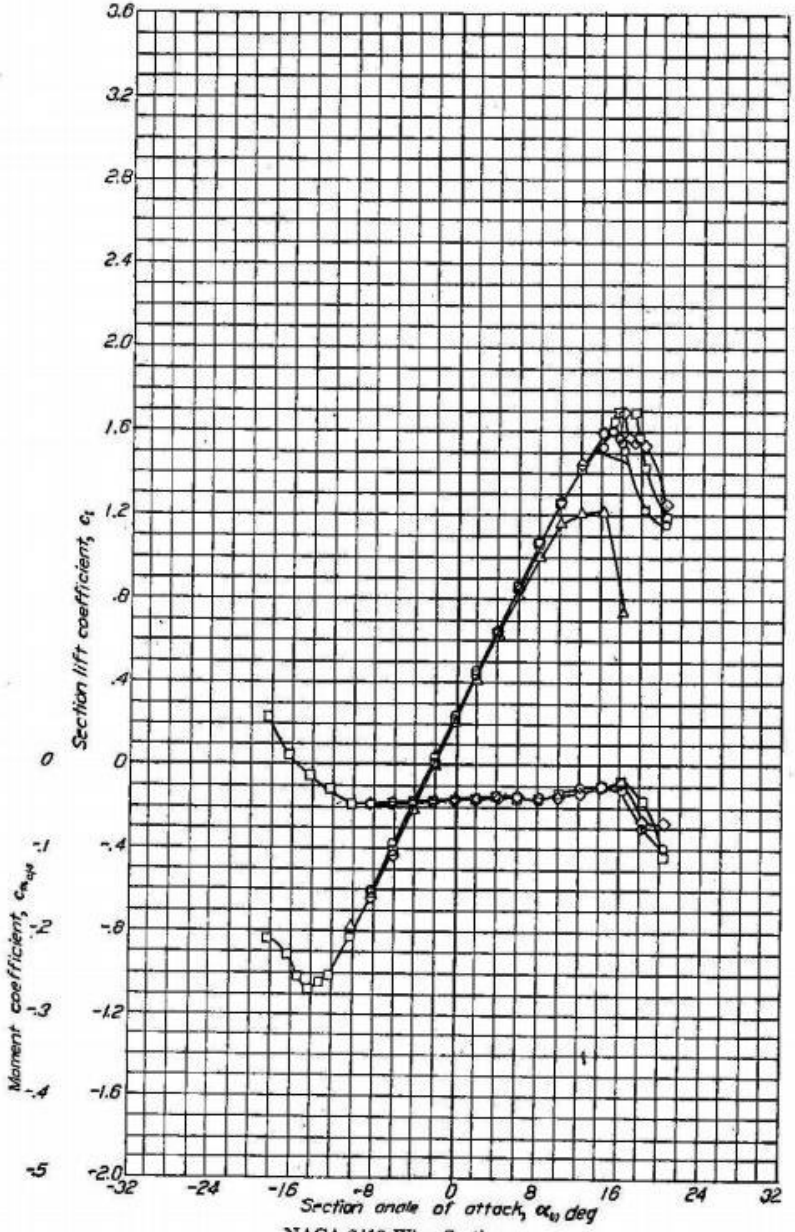
$$Lift = C_L \bar{q} S = C_L \left(\frac{1}{2} \rho V^2 \right) S$$

$$Drag = C_D \bar{q} S = C_D \left(\frac{1}{2} \rho V^2 \right) S$$

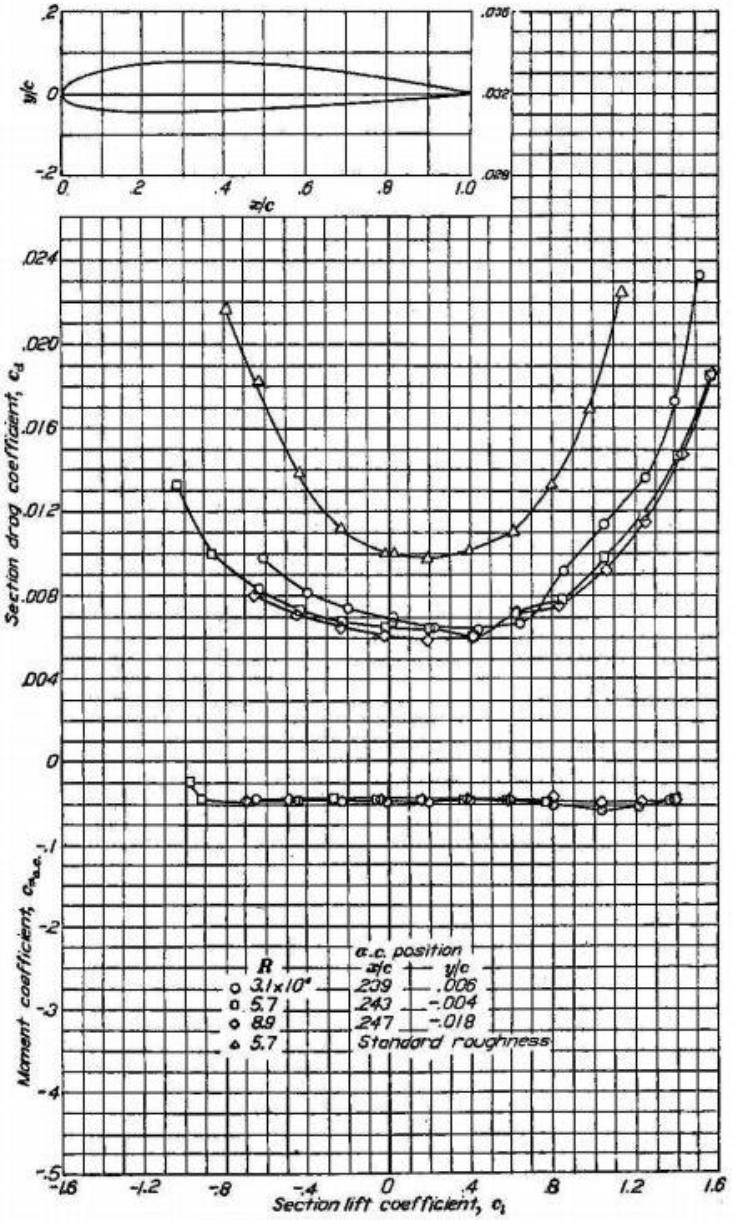
Lift Coefficient and Drag Coefficient

- The lift coefficient, C_L and Lift L_w are
  - C_L = Lift/Impact force = L_w/QS
  - $L_w = (1/2\rho V_T^2 S)C_L$

- The drag coefficient, C_D and Drag D_w are
  - C_D = Drag/Impact force = D_w/QS
  - $D_w = (1/2\rho V_T^2 S)C_D$



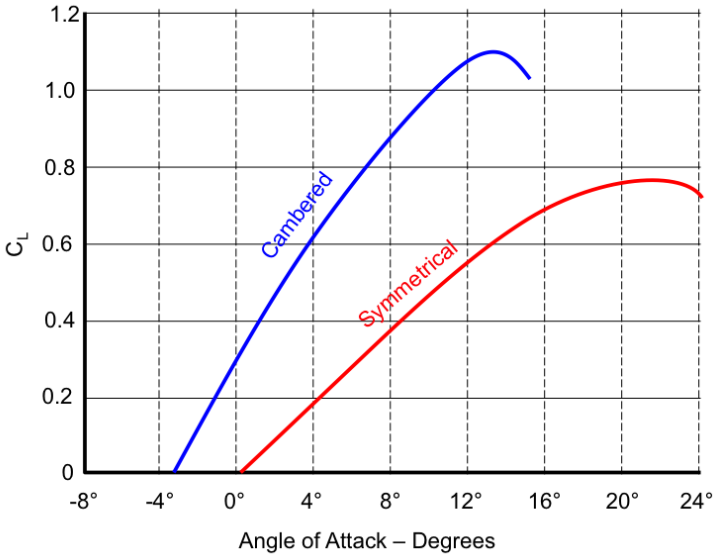
NACA 2412 Wing Section

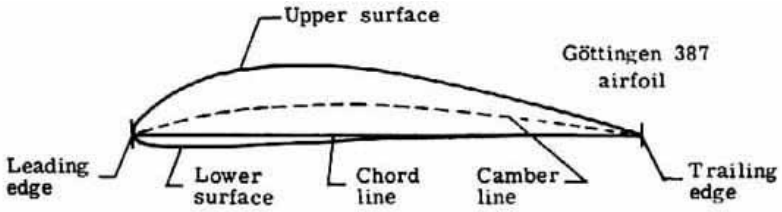


NACA 2412 Wing Section (*Continued*)

Aerodynamic Lift

- 2-D Lift (inviscid, incompressible flow)





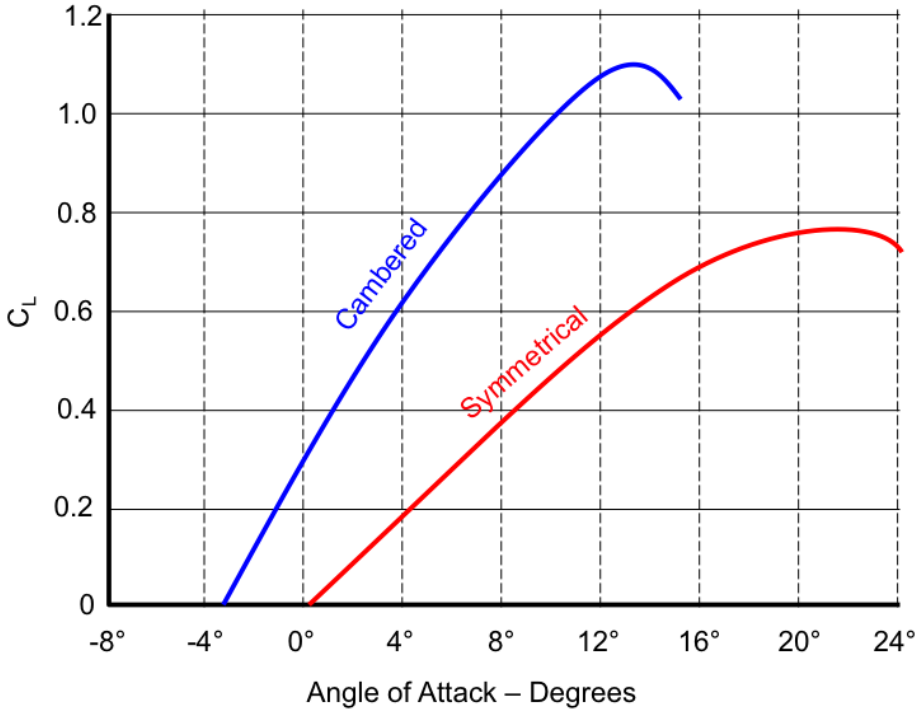
$$(\mathbf{Lift})_{2\text{-}D} \simeq \left[\frac{1}{2}\rho_\infty V_\infty{}^2 c\left(C_{L_\alpha}\right)_{2-D}\alpha\right] + \left[\rho_\infty V_\infty \left(\Gamma_{camber}\right)_{2-D}\right]$$

$$= \textbf{[Lift due to angle of attack]}$$

$$+ \textbf{[Lift due to camber]}$$

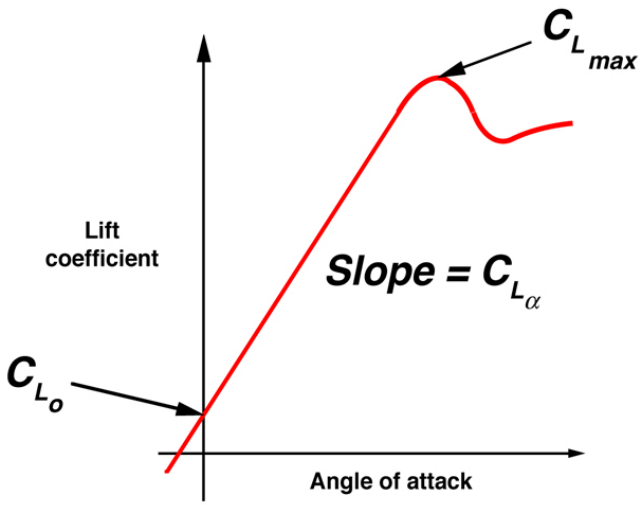
Airfoil to Airflow

- Airfoil to airflow has always an angle to be controlled to change the interacting force on the wing
- AoA increases, lift increases in general
- Airfoil cross section changes, airflow structure is changed to quickly drop the lift.



Aerodynamic Lift

- Lift vs. Angle of Attack



- For free-stream,

$$L = f(V_\infty, \rho_\infty, S, \mu_\infty, a_\infty)$$

- 3-D Lift,

$$Lift = C_L \frac{1}{2}\rho V_{air}^2 S \approx \left(C_{L_{wing}} + C_{L_{fuselage}} + C_{L_{tail}}\right)\frac{1}{2}\rho V_{air}^2 S \approx \left[C_{L_0} + \frac{\partial C_L}{\partial \alpha}\alpha\right]\bar{q}S$$

Affecting Lift

- (1) Lift coefficient C_L (function of α)
  - C_L is determined by airfoil design by upper curvature
  - Extract Flap and Slat to increase wing area and curvature to increase lift
  - Full flap and slat contributes 100% C_L
  - Low speed during takeoff and landing, lift $L=C_L qS$, *q is small, L* can only be increased by changing C_L
  - After takeoff, the speed increases to get larger q, flap and slat are retracted to avoid drag in high speed.
  - Flap and slap being retracted, C_L drops and also drag drops.
  - For a given airfoil,

$$c_l = f(\alpha, M_\infty, \mathrm{R_e})$$

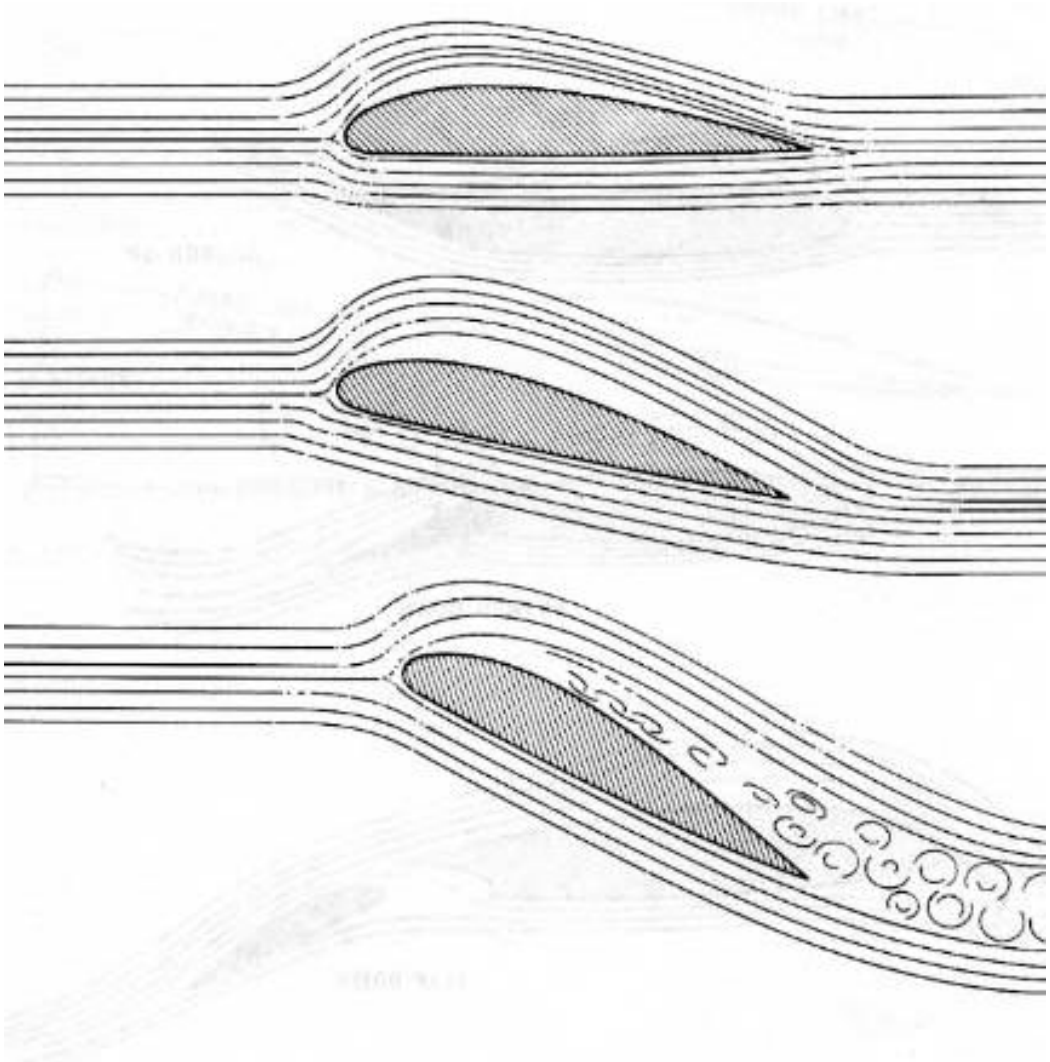
Affecting Lift

- (2) Dynamic Pressure q
  - Lift is proportional to q, $q = 1/2\rho V_T^2$ ，q is also proportional to speed, air density
  - Low air flight has large drag to consume high fuel
- (3) Wing area
  - Larger wing area to get larger lift (but larger drag a well)
  - Aircraft at low speed may extend wind area by flap and slat

Airfoil to Airflow

- Low AoA
- Higher AoA
- Over AoA

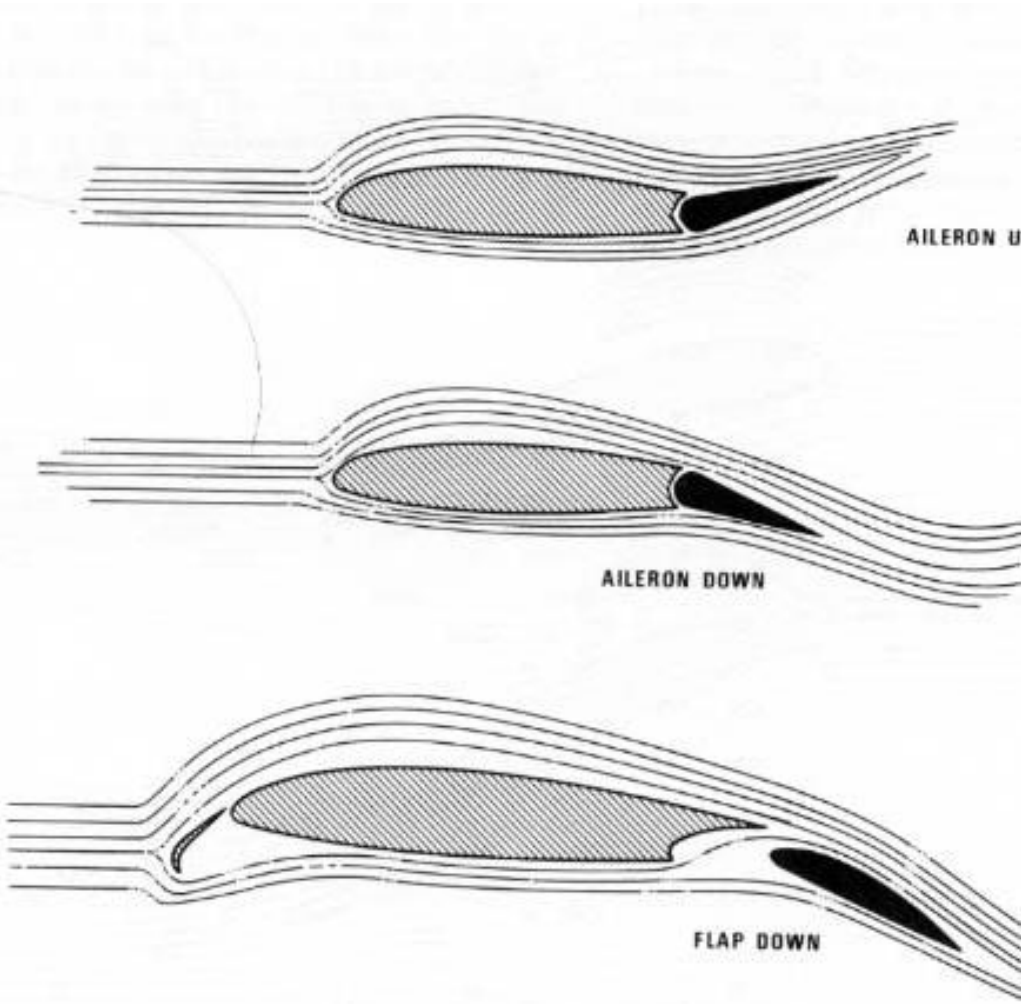
Flow has been spoiled to vanish lift



Flow Change from Wing Change

- Aileron up
- Aileron down
- Flap extends

  (Increase wing area)



Relationship of Lift to Drag

- Impact pressure applies on wings to divide into axial component of drag and perpendicular component as lift
- Larger wing results in larger impact force Q, and also larger drag
- Drag is $D_w = QSC_D$
- Lift coefficient and drag coefficient is
  - $C_D = C_{D0} + K\,C_L^2$
  - C_{D0} and K are coefficients when wing is designed

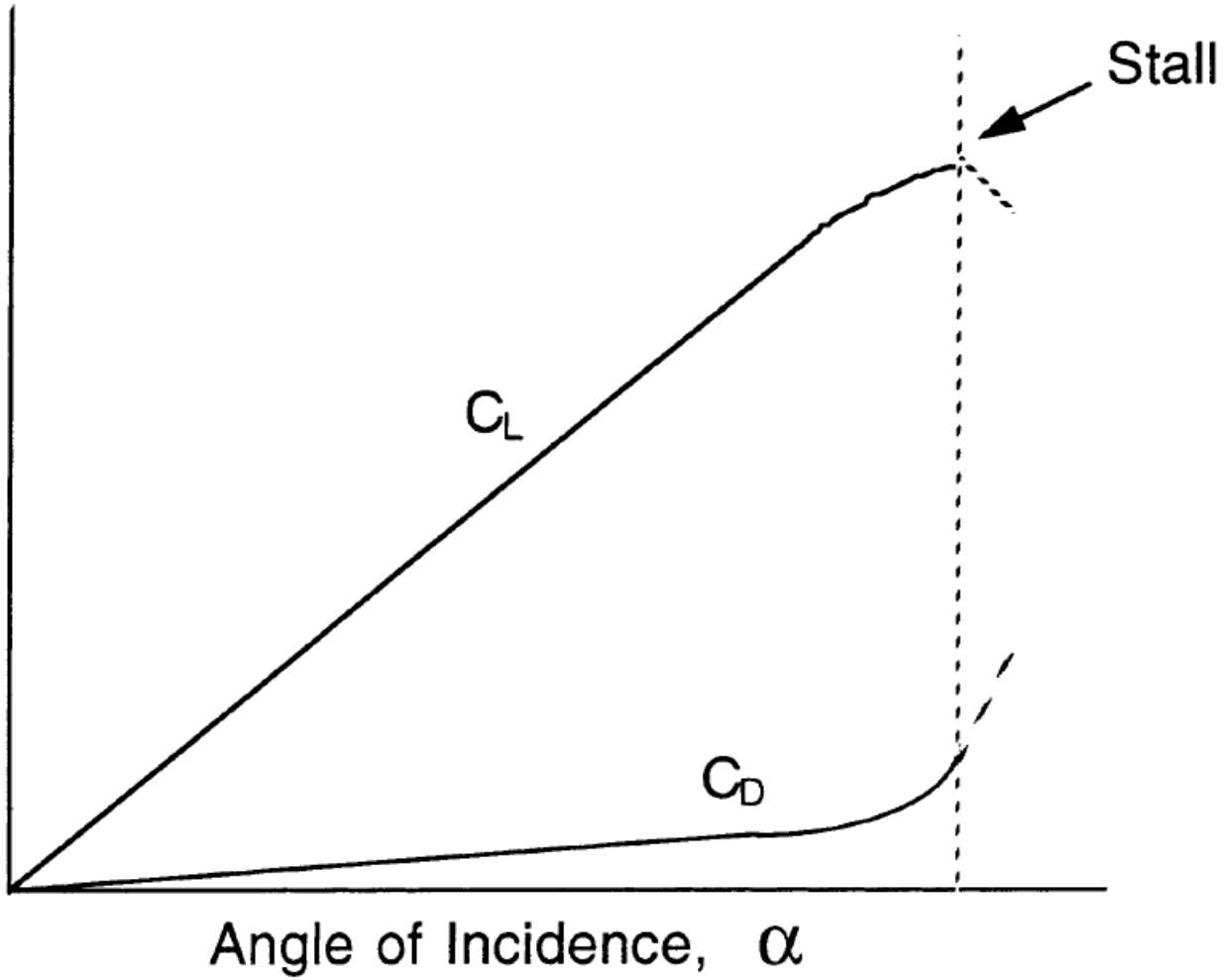
Relationship of Lift to Drag

- C_D is proportional to C_L^2, when C_L is small, it is even smaller. But when C_L>1, drag increases fast.
- For most AoA, C_D is relatively small. Only near stall angle, drag increases drastically.
- Therefore, AoA contributes to lift, and little drag under normal maneuvering.

AoA to Stall

- Civil aircraft designs AoA below 15 ～25 °
- Military aircraft may design from 25 ~35° for high performance
- When AoA increases to a certain limit, drag increases quickly and accompanying a drastic lift drop. This phenomenon is called “Stall”.
- High AoA may also decrease air inflow to engine, and reduce engine thrust.

AoA, Lift and Drag



Effect of Flap to Lift

- Camber modification
- Trailing-edge ***flap*** deflection shifts C_L up and down
- Leading-edge flap (***slat***) deflection increases stall α
- Same effect applies for other control surfaces
  - Elevator (horizontal tail)
  - Ailerons (wing)
  - Rudder (vertical tail)

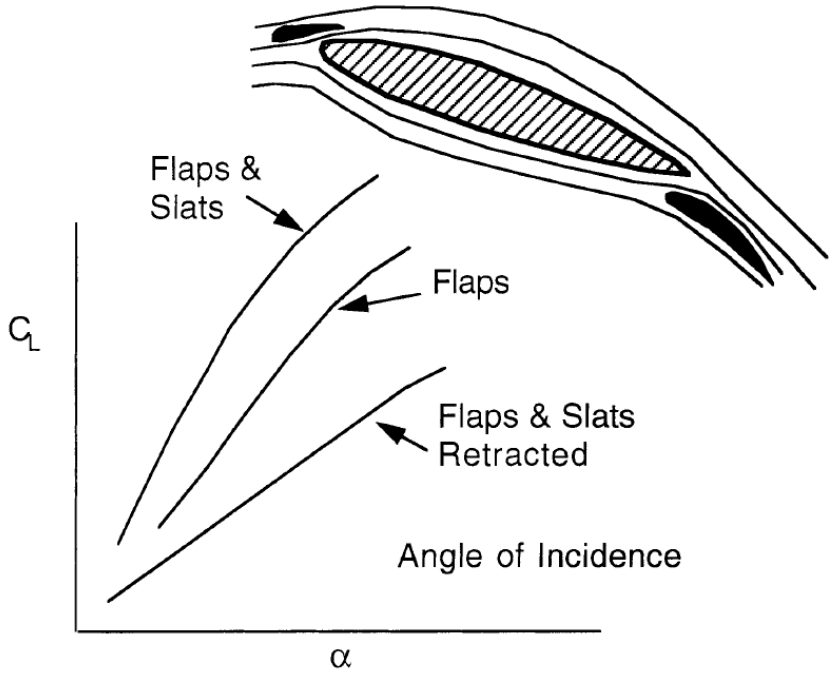


Fig. 3.5 Leading edge slats and trailing edge flaps.

Illustrative Example on Basic Aerodynamics

- A simple example of the application of these basic aerodynamic relationships is set out below.
- Air data Air density, $\rho 0$, at sea level and standard temperature (15°C) and pressure (1013.25 mbar) is 1.225 kg/m^3.
  Air density, ρ, at 30,000 ft (9,144 m) is 0.4583 kg/m^3.

Aircraft mass, m	30,000 kg
Wing area, S	75 m^2
Maximum lift coeff, $C_{L\,\max}$	1.2
Maximum angle of incidence, $\alpha_{\max}$	15°

C_L versus α relationship is assumed to be linear.

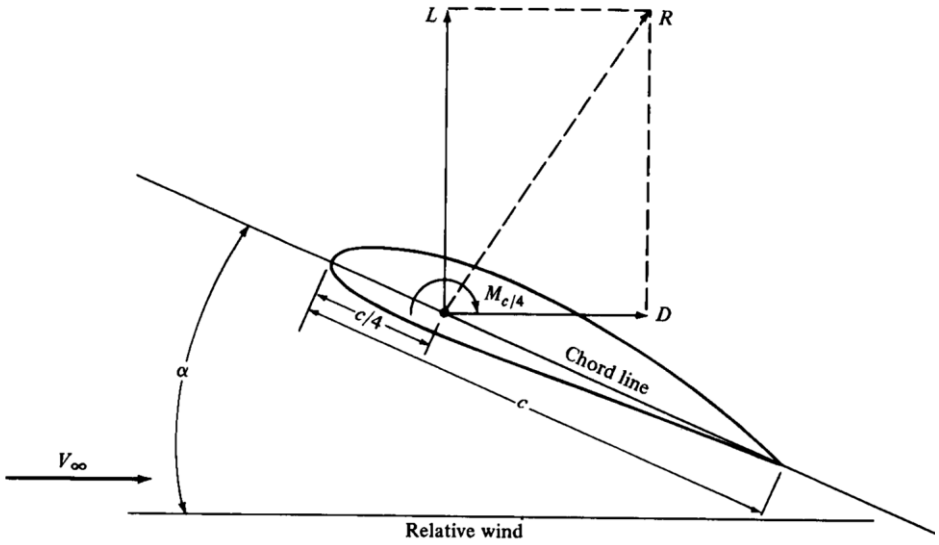
Illustrative Example on Basic Aerodynamics

- *Question (i)* What is the wing incidence, α, when flying straight and level at a speed of 80 m/s (160 knots approx.) at a height of 200 ft in the approach to the airfield?
- *Question (ii)* What is the corresponding wing incidence when flying straight and level at a speed of 200 m/s (400 knots approx.) at a height of 1,000 ft?

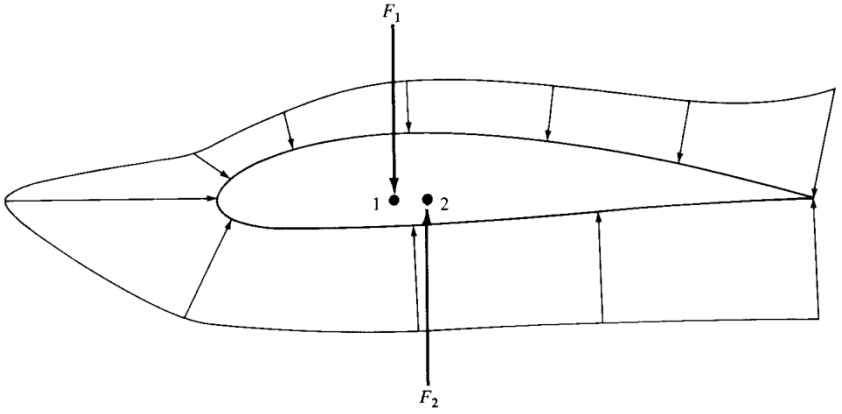
  Note: Neglect change in density 0–1,000 ft for simplicity.
- *Question (iii)* What is the maximum normal acceleration that can be achieved at a height of 30,000 ft when flying at a speed of 225 m/s (450 knots)? Neglect compressibility effects and assume adequate thrust is available to counteract the increase in drag and maintain speed of 450 knots at α_{max}.

Moments of an Airfoil

- In addition to lift and drag, the surface pressure and shear stress distributions create a moment **M** which tends to rotate the wing.
- To see more clearly how this moment is created, consider the surface pressure distribution over an airfoil.
- Dynamic pressure causes pitch moments to change aircraft longitudinal attitude.



Sketch showing the definitions of (*b*) normal and axial force.



The physical origin of moments on an airfoil.

Moments of an Airfoil

- Moment coefficient about the Leading edge

$$M_{LE} = -L\left(\frac{c}{4}\right) + M_{c/4}$$

$$\frac{M_{LE}}{q_\infty Sc} = -\frac{L}{q_\infty Sc}\left(\frac{c}{4}\right) + \frac{M_{c/4}}{q_\infty Sc}$$

$$\boxed{c_{m_{LE}} = -\frac{c_\ell}{4} + c_{m_{c/4}}}$$

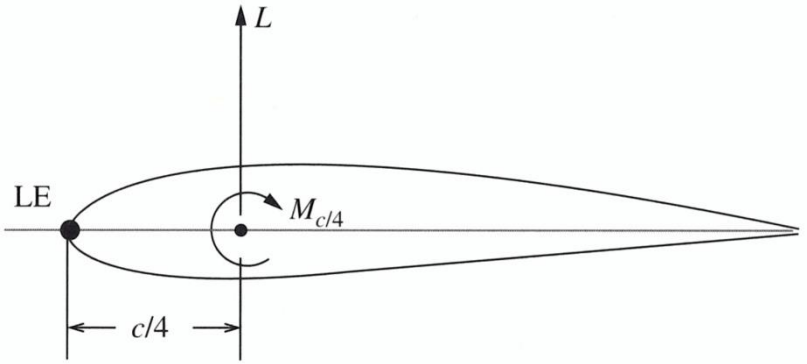


Figure 5.11 Sketch of lift and moments on an airfoil.

Aerodynamic Reference Centers

- Airfoil with centers
  - By placing all of the lift and drag on the aerodynamic center we move the lift and drag due to camber from the half-chord to the quarter chord.
  - This pitching moment or couple experienced at zero lift, M_0, is again expressed in terms of a non-dimensional coefficient, pitching moment coefficient, M_0

$$C_{M_0} = \frac{M_0}{\frac{1}{2}\rho {V_T}^2 Sc}$$

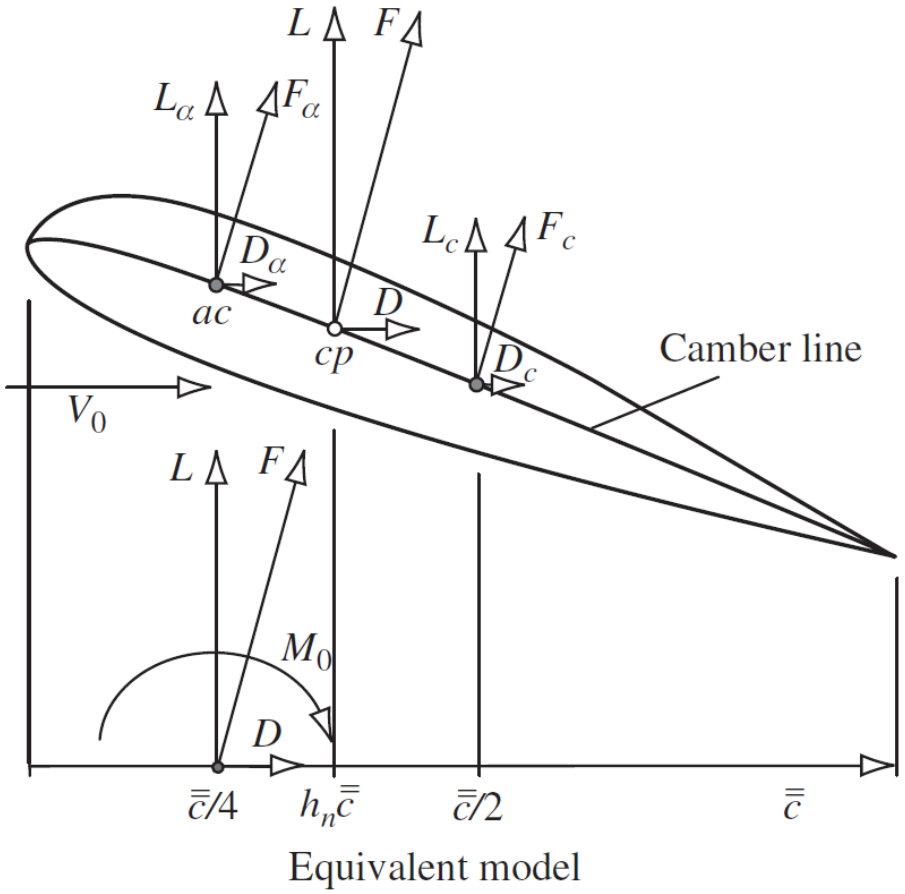
  - where c is the *mean aerodynamic chord*, equal to wing area/wing span. Thus, pitching moment

$$M_0 = \frac{1}{2}\rho V_T^2 Sc\ C_{M_0}$$

Aerodynamic Reference Centers

- Center of pressure (*cp*): The point at which the resultant aerodynamic force *F* acts. There is no aerodynamic moment around the *cp*.
- Half-chord: The point at which the aerodynamic force due to camber, F_c, acts
- Quarter-chord (or aerodynamic center): The point at which the aerodynamic force due to angle of attack, F_a, acts. The aerodynamic moment around the quarter-chord, M_0, is constant with angle of attack.
  At supersonic speeds it tends to move aft to the half chord point.

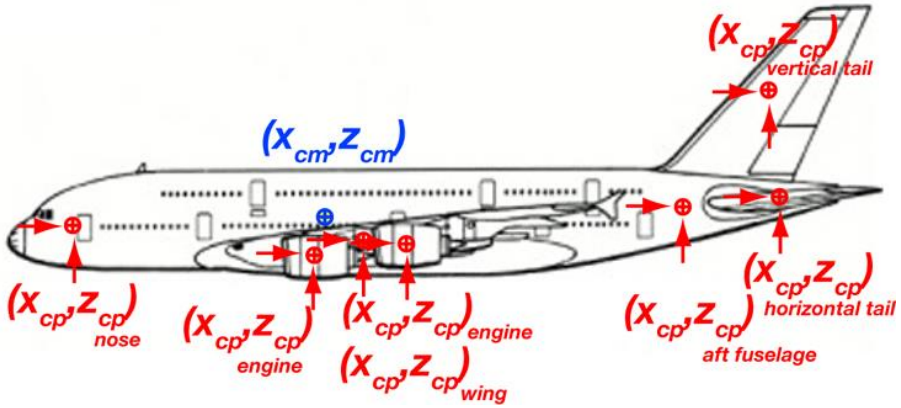
Aerodynamic Reference Centers



Aerodynamic reference centers.

Pitching Moment of the Airplane

- Pitching moment is moment about the y axis of the body frame
  - Distributed effects can be aggregated to local centers of pressure

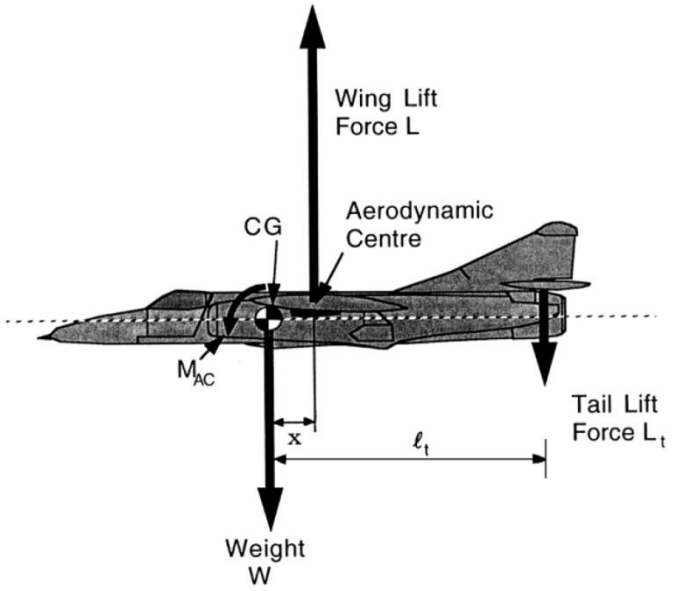


- **Net effect expressed as**

$$M_B = C_m \bar{q} S \bar{c}$$

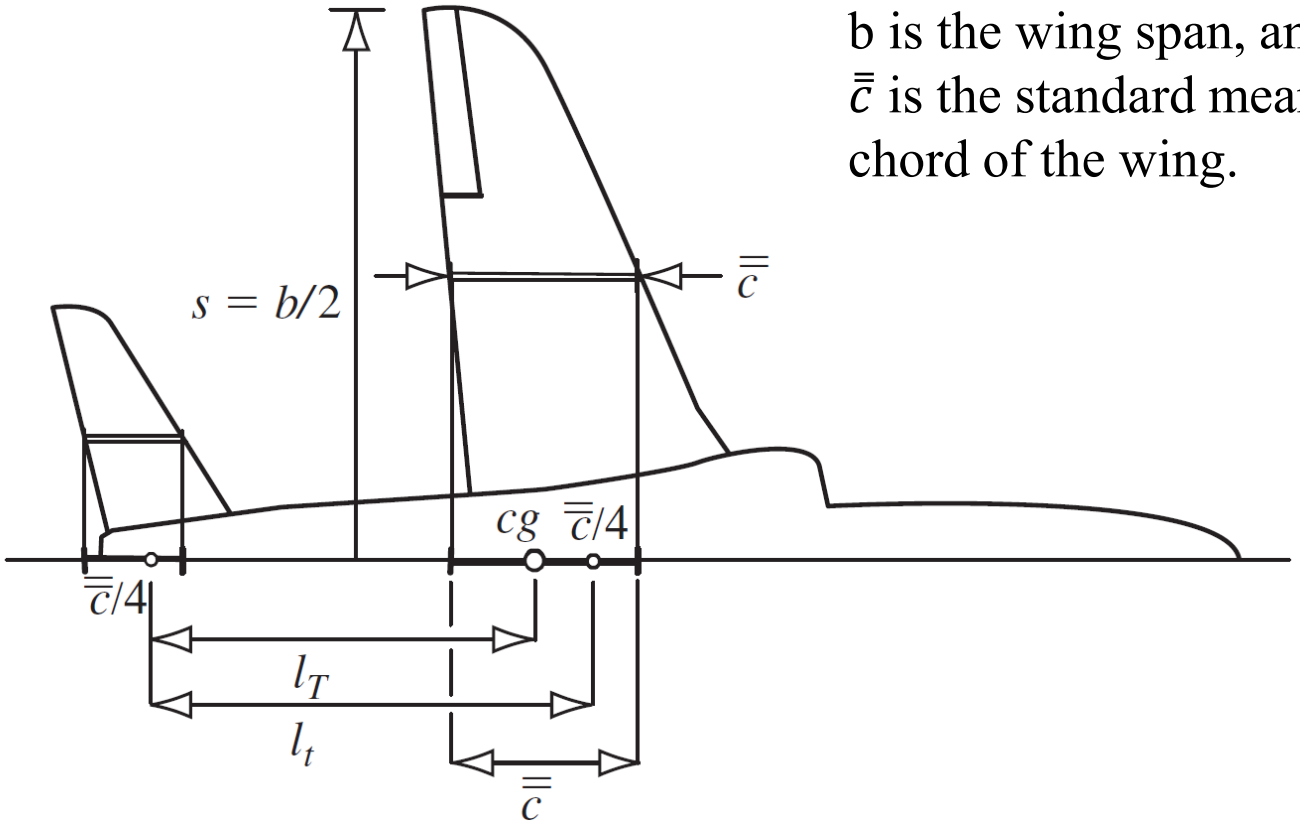
Pitching Moment of the Airplane

- Aerodynamic moments (or Torques)
- Airplane balance for conventional aft-tail configuration
  - cg near wing's aerodynamic center
  - aerodynamic center: the point at which wing's pitching moment coefficient is invariant with angle of attack ~25% mean aerodynamic chord (MAC)



Aircraft Reference Geometry

- Longitudinal reference geometry



b is the wing span, and $\bar{\bar{c}}$ is the standard mean chord of the wing.

Longitudinal reference geometry.

Pitching Moment of the Airplane

- Both lift and drag will cause a force moment on the fuselage.
- Zero lift pitch moment M_0 is the portion from Pitch moment with no action to lift
- The location of zero moment on aircraft fuselage is called center of pressure, other locations are not.
- It is pitch moment describing by pitch moment coefficient as
  - $C_{Mo} = M_0/(1/2\rho V_T^2 S\bar{c})$

Tailplane Contribution

- The total pitching moment about the aircraft's CG and its relationship to the AoA is of prime importance in determining the aircraft's stability.
- The tailplane makes a major contribution to longitudinal stability and provides the necessary downward lift force to balance or trim the aircraft for straight and level flight.

AoA Trim to Pitch Moment

- For maintaining altitude cruise, the external force onto the aircraft should get zero total moment.
- When AoA changes, pitch moment changes to affect flight stability.
- To flight stability, trim angle of incidence(α_T) is defined relatively to C_M from zero AoA
- When AoA is larger or smaller than α_T, a reverse response of C_M appears to maintain aircraft in stable flight.

C_M, α_T and Stability

- Resultant moment about CG, M, is given by

$$M = -L_W x - M_0 + L_t l_t \qquad (3.9)$$

(nose-up moments are defined as positive).

Wing lift:

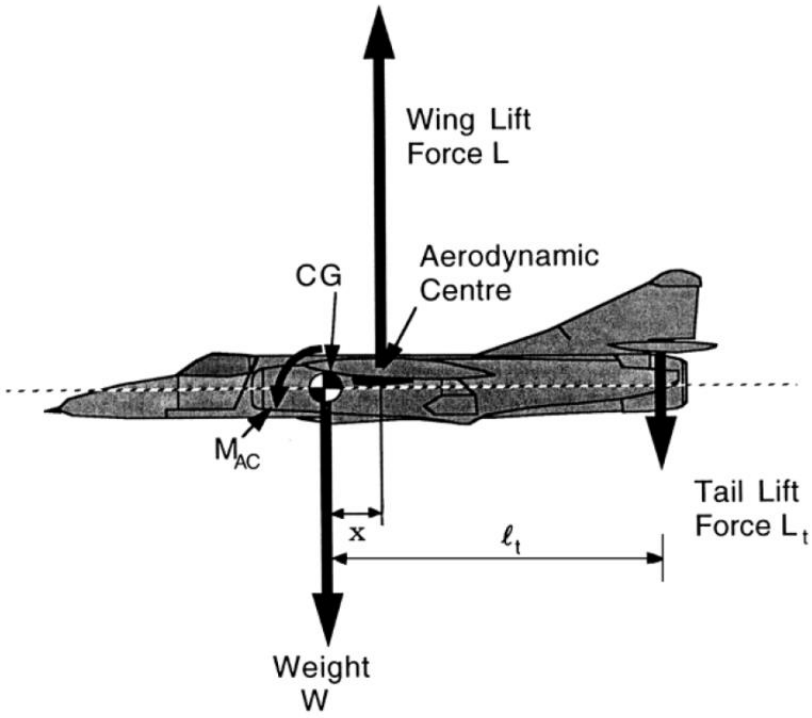
$$L_W = \frac{1}{2}\rho V_T{}^2 S\, C_L$$

- Wing pitching moment:

$$M_0 = \frac{1}{2}\rho V_T{}^2 Sc\; C_{M_0}$$

- Tailplane lift:

$$L_t = k_t \frac{1}{2}\rho V_T{}^2 S_t\, C_{Lt}$$



C_M, α_T and Stability

- Dividing both sides of equation (3.9) by $(1/2)\rho V_T^2 Sc$ yields

$$C_M = -\frac{x}{c}C_L - C_{M_0} + k_t \frac{(S_t l_t)}{Sc} C_{Lt} \qquad (3.12)$$

- That is, $dC_M/d\alpha$ is more negative and this increases the aerodynamic stability

$$\frac{dC_m}{d\alpha} < 0$$

C_M, α_T and Stability

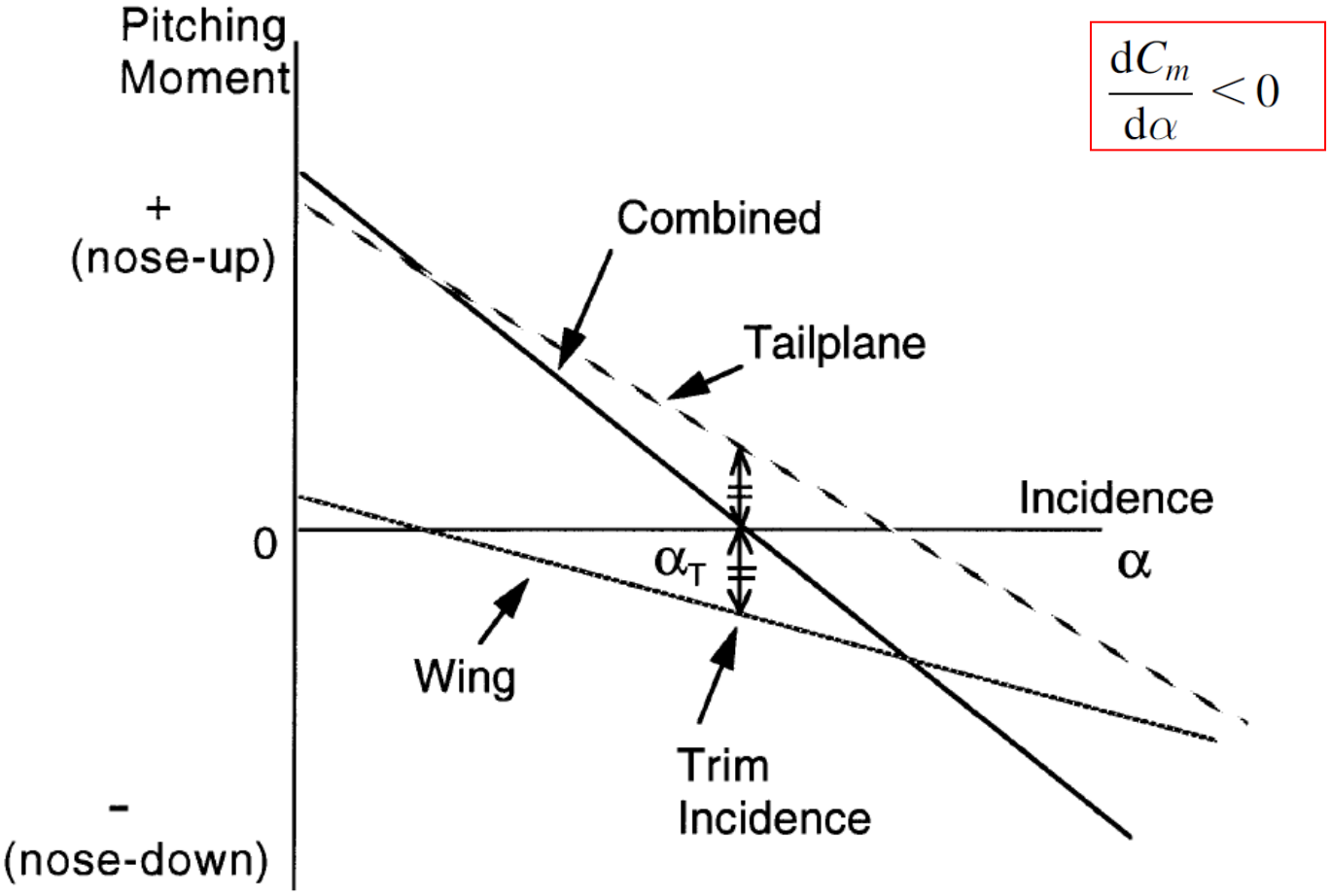


Fig. 3.7 Pitching moment versus incidence.

Aircraft Stability

- Stable and unstable happens in daily life

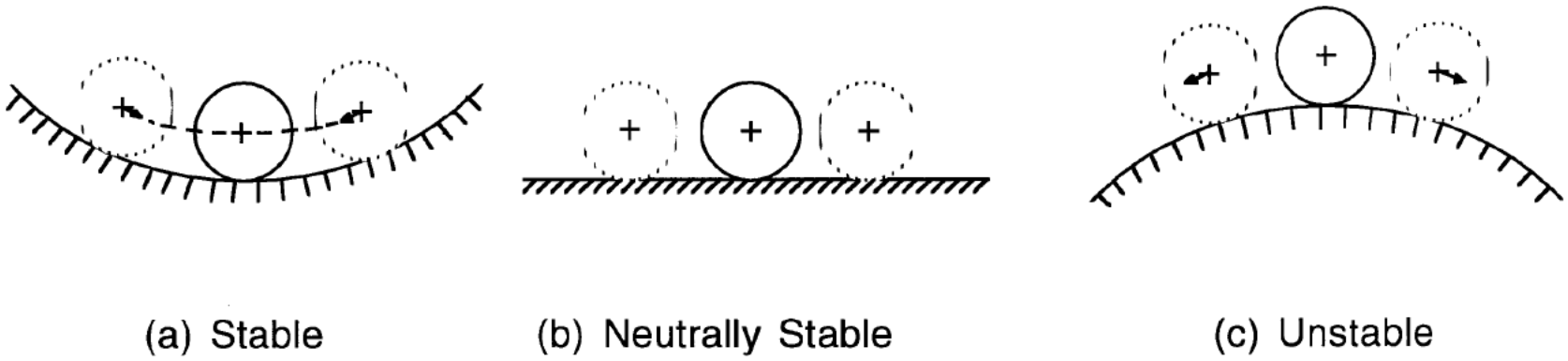
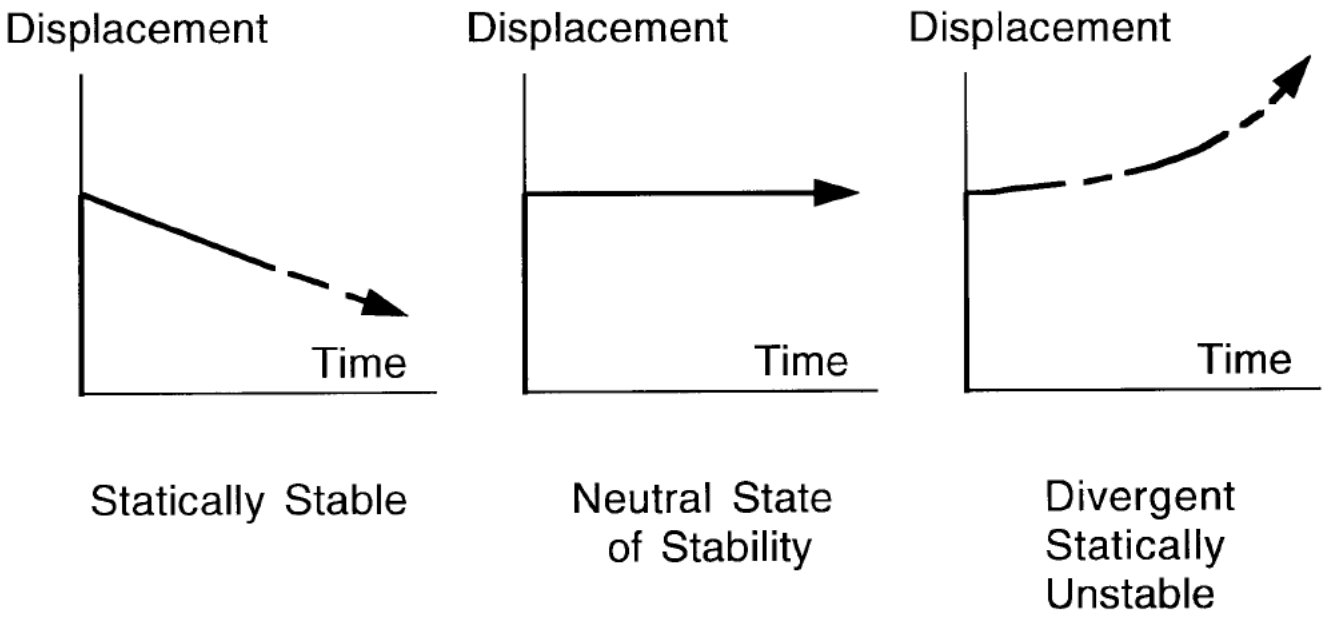


Fig. 3.8 Simple example of stability conditions

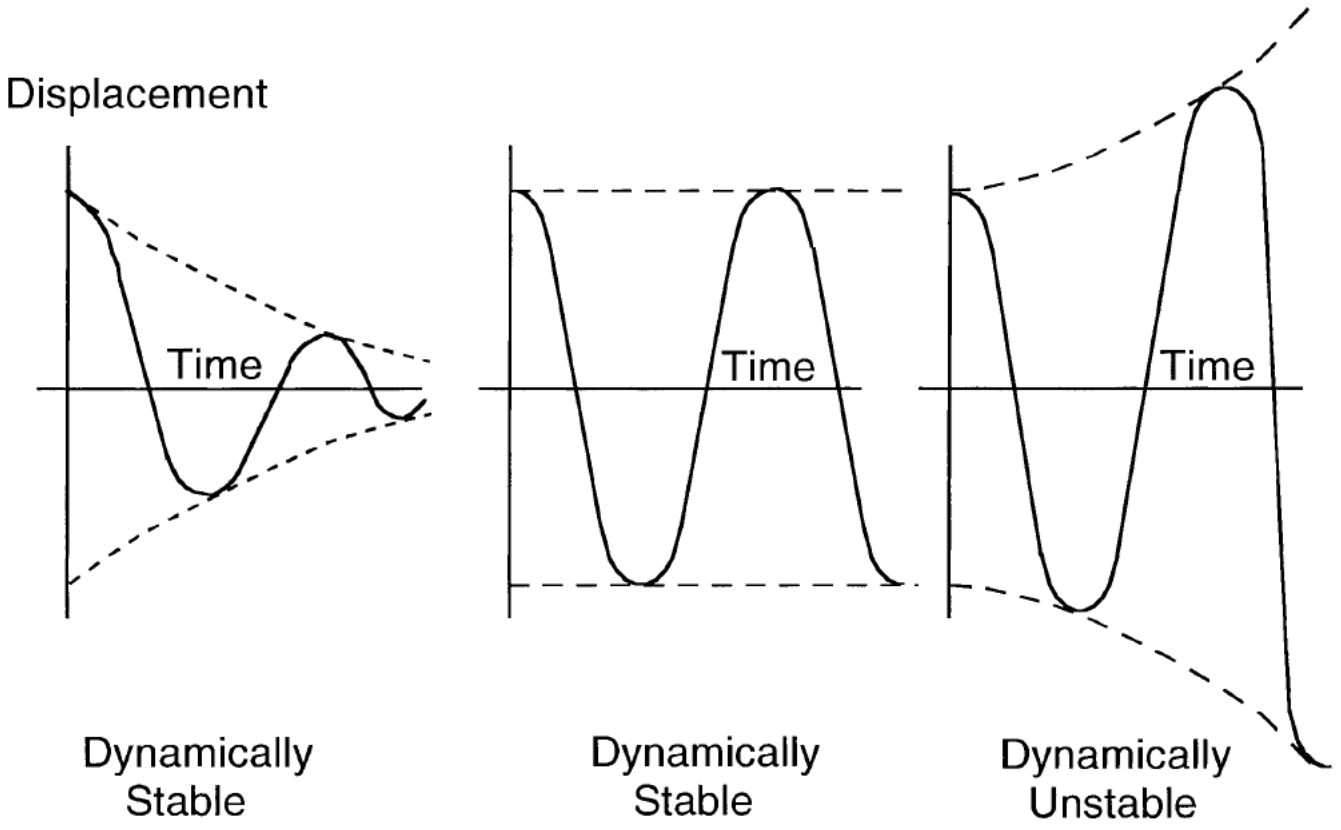
Longitudinal Stability

- Any final result goes to a target means stable



Longitudinal Stability

- Any final result is bounded within a range means stable, otherwise goes to infinity

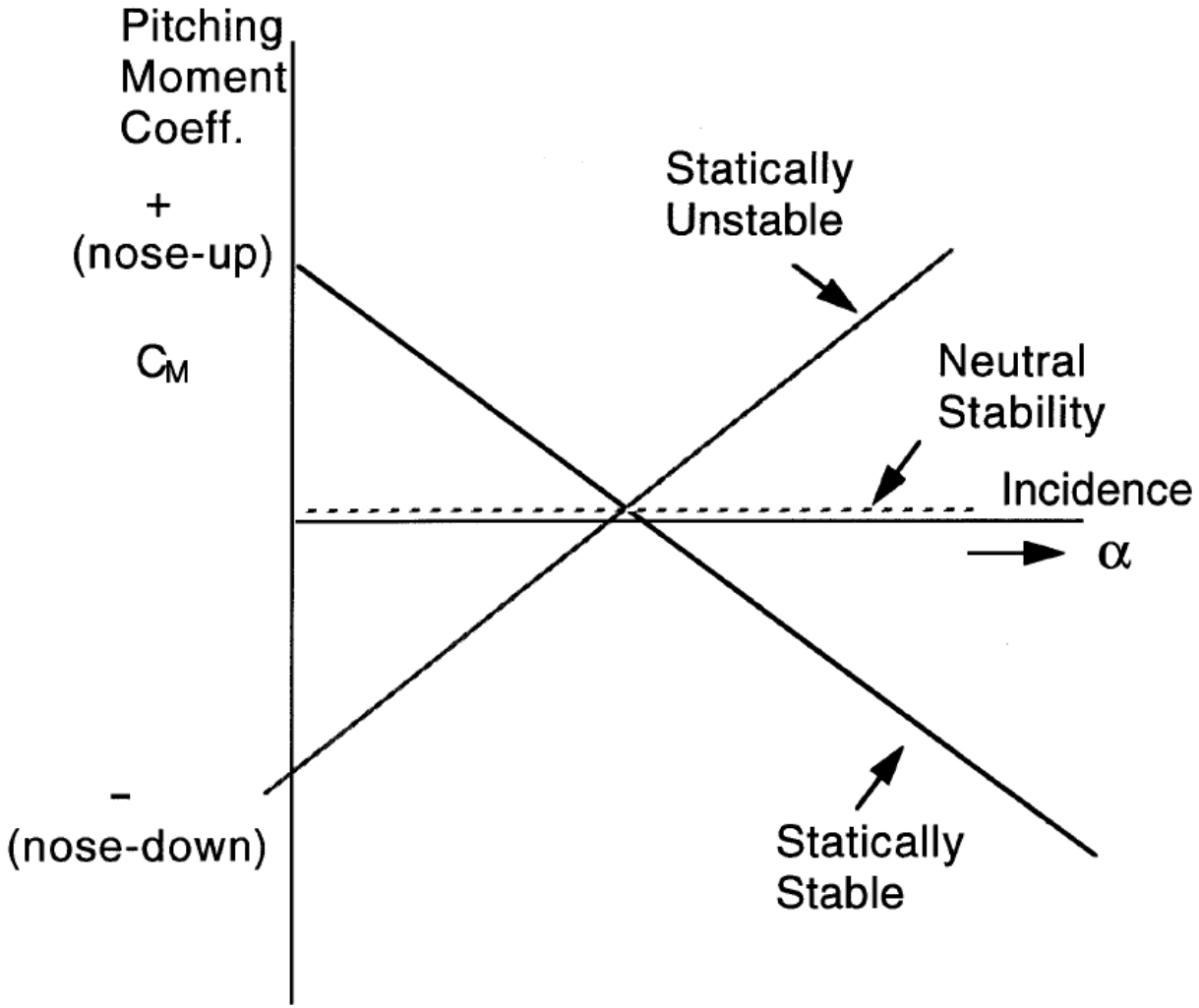


Conditions for Stability

- The necessary criteria for longitudinal balance and static stability are
  - 1 . $\boldsymbol{C_{M,0}}$ must be positive.
  - 2. $\partial C_{M,cg}/\partial \alpha_a$ must be negative.
  - 3. $\boldsymbol{C_M} = 0$ at $\alpha = \alpha_T$
- Since α and $\boldsymbol{C_L}$ are linearly, or very nearly linearly, $\boldsymbol{C_L}$ ***is*** related by the lift curve slope ***a***.
- The rather more complex analysis by Gates and Lyon (1944) takes speed effects into account and defines a general requirement for longitudinal static stability as

$$\frac{\mathrm{d}C_m}{\mathrm{d}C_L} < 0$$

C_M versus α

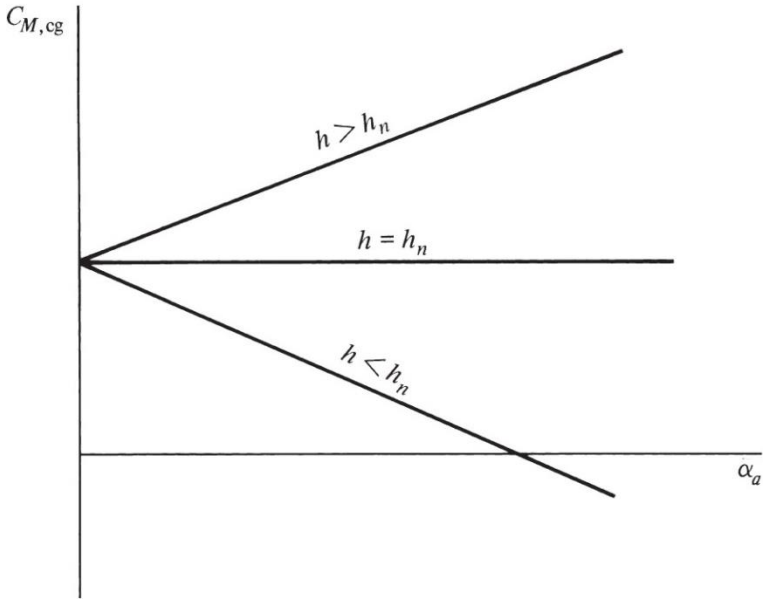


Neutral point (NP)

- **Neutral point (NP)** is one specific location of the center of gravity such that

$$\frac{\partial C_{M,cg}}{\partial \alpha_a} = 0$$

- The value of ***h*** when this condition holds is defined as the ***neutral point,*** denoted by h_n. When $h = h_n$, the slope of the moment coefficient curve is zero



Static Margin

- **Static margin *(SM)* reflects the distance between the center of gravity *(cg)* and the net center of pressure *(cp)***
  - **Body axes**
  - **Normalized by mean aerodynamic chord**
  - **Does not reflect *z* position of *center of pressure***
- **Positive *SM* if *cp* is behind *cg***

$$\text{Static Margin} \triangleq SM = \frac{100\left(x_{cg} - x_{cp_{net}}\right)_B}{\bar{c}}, \%$$

$$\equiv 100\left(h_{cg} - h_{cp_{net}}\right)\%$$

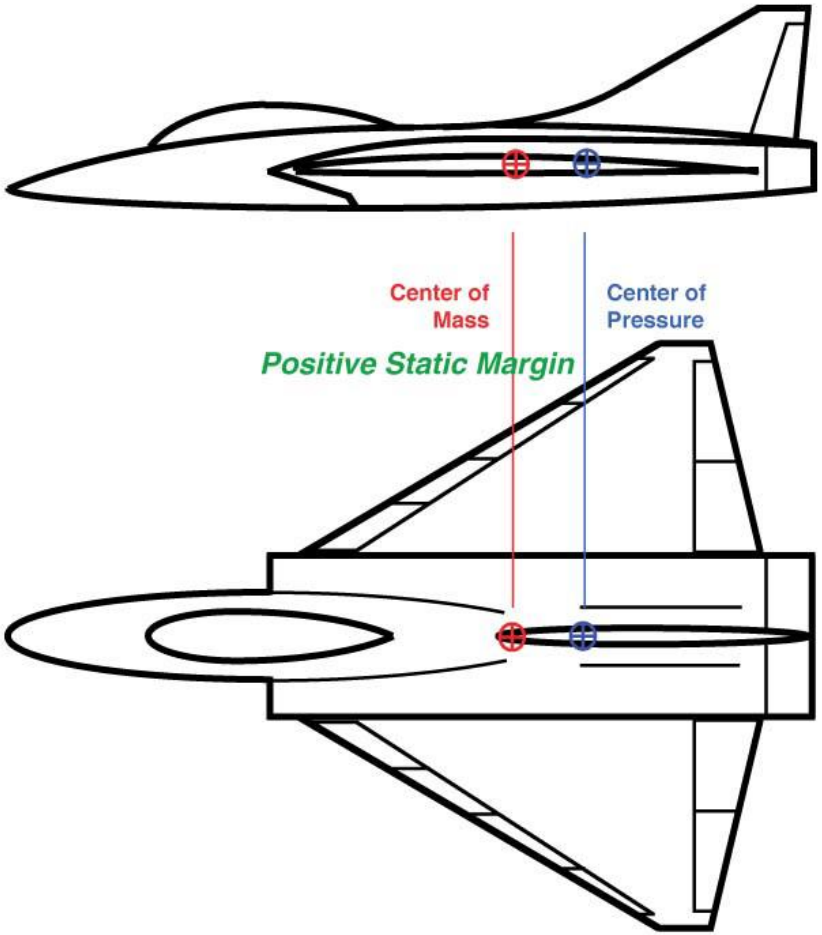
$$h_{cm} \triangleq \frac{x_{cg}}{\bar{c}}$$

Static Margin

$$\textit{Static Margin} \triangleq SM$$

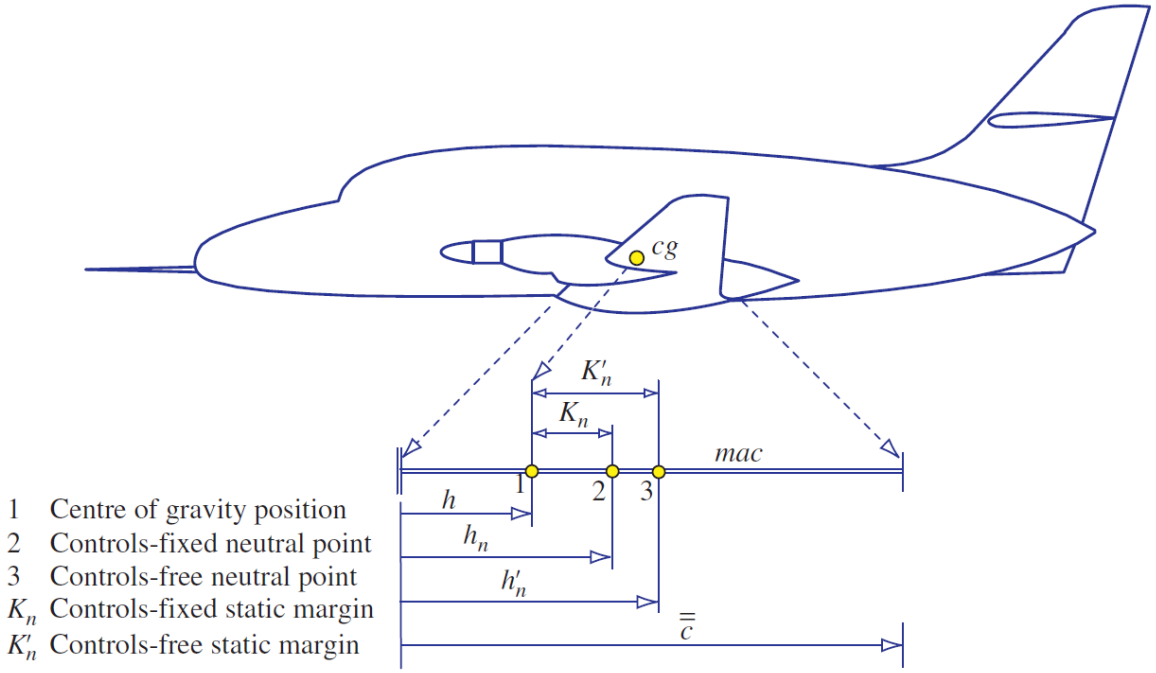
$$= \frac{100\left(x_{cg} - x_{cp_{net}}\right)_B}{\bar{c}}, \%$$

$$\equiv 100\left(h_{cg} - h_{cp_{net}}\right)\%$$



Summary of Longitudinal Static Stability

- The important parameters are **neutral point (NP)** positions and their relationship to the ***cg*** position, which in turn determines the **stability margins (SM)** of the aircraft.
- The stability margins determine literally how much stability the aircraft has in hand, in the controls- fixed and controls-free senses, over and above neutral stability. The margins therefore indicate how safe the aircraft is.



Longitudinal stability margins.

Aerodynamic Instability

- When airflow on wings cannot offer natural stability and balance is termed as aerodynamic instability
- Aerodynamic instability may be controlled into extreme and critical performance
- For simplicity and flight safety, high performance and precision flight control should be applied to maintain flight stability

Aerodynamic Instability

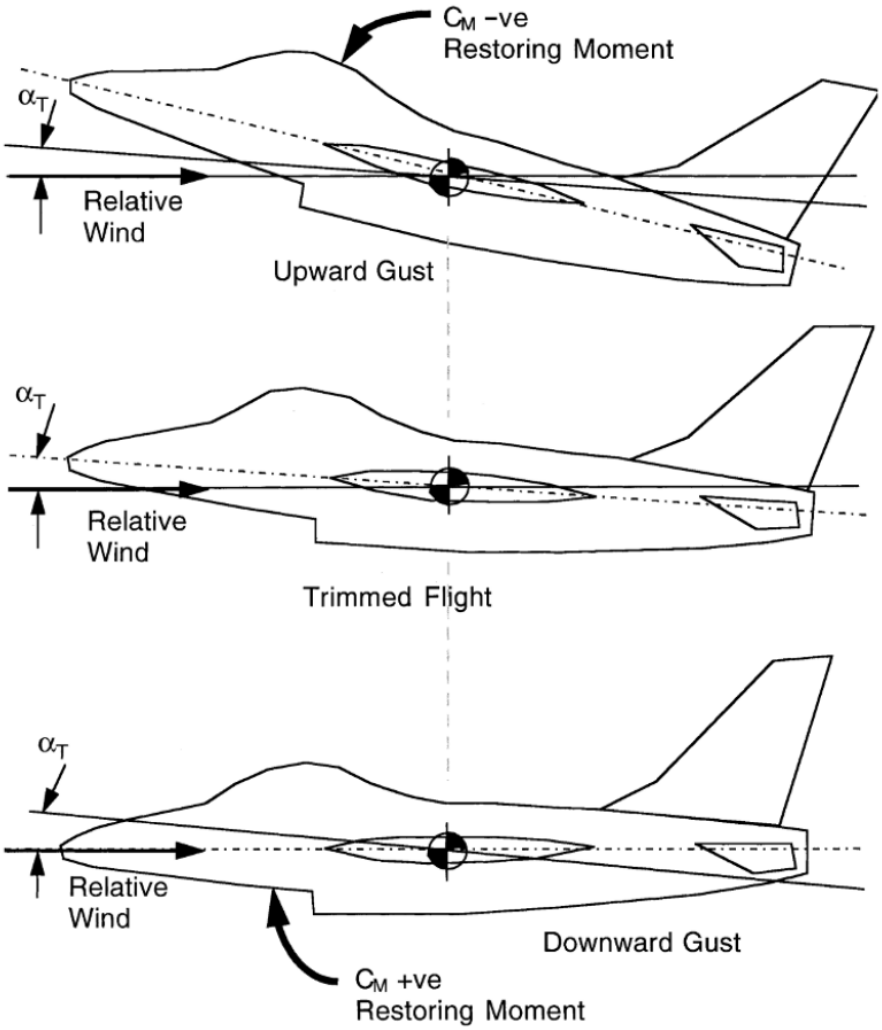


Fig. 3.11 Natural longitudinal stability.



X-29 Forward-swept wing

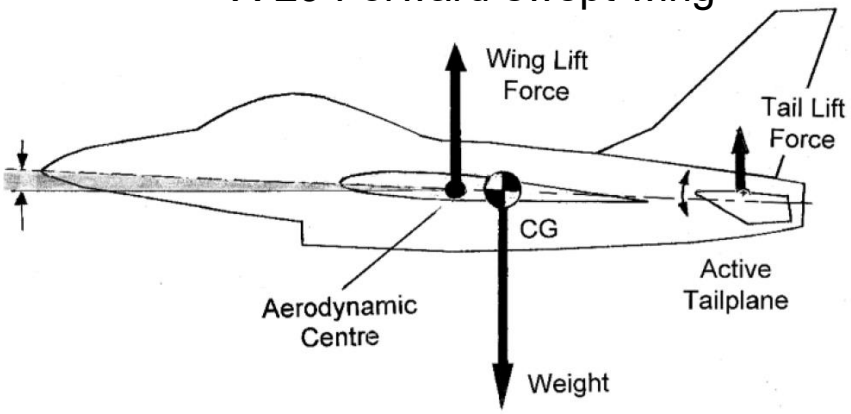
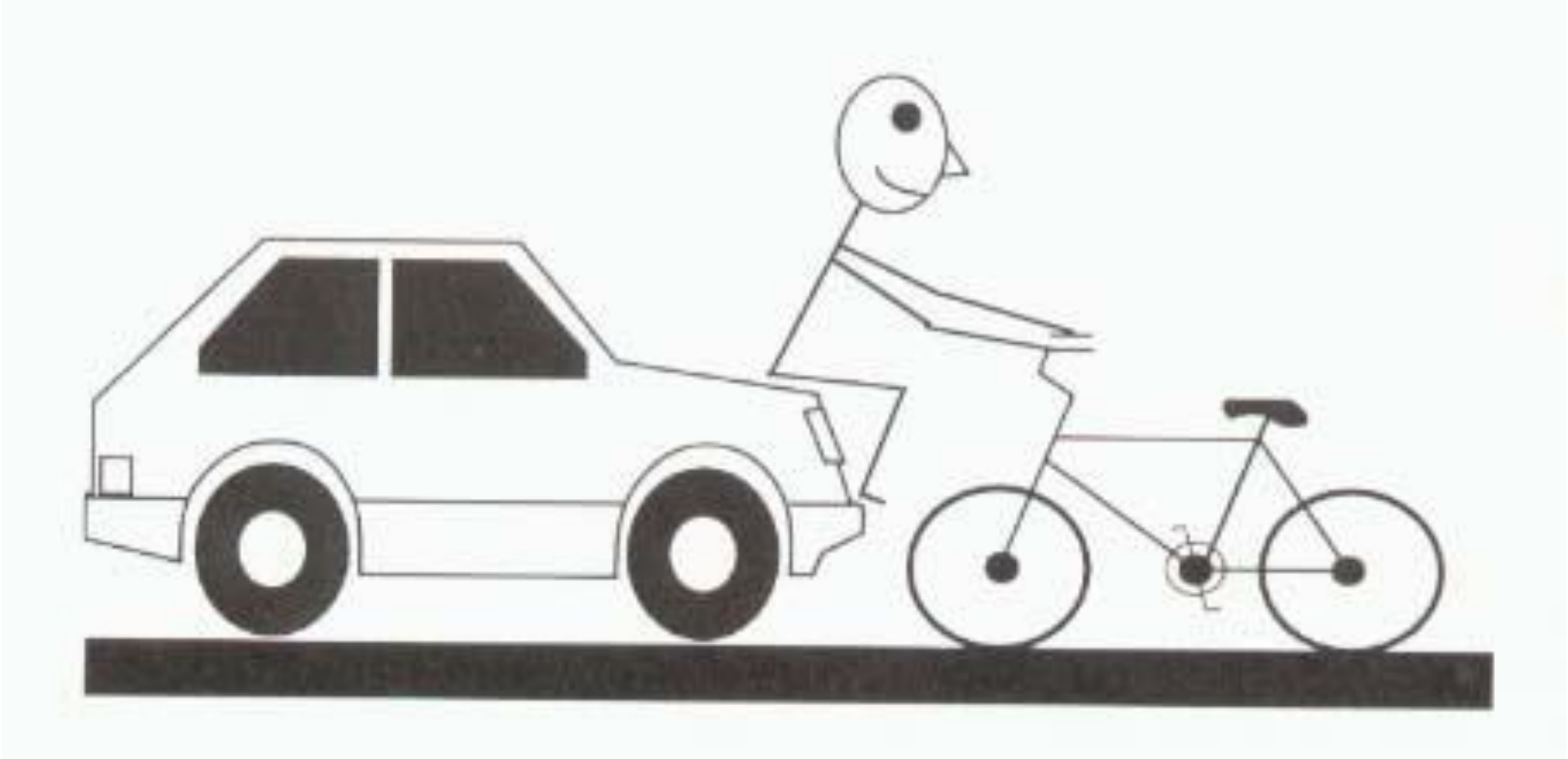
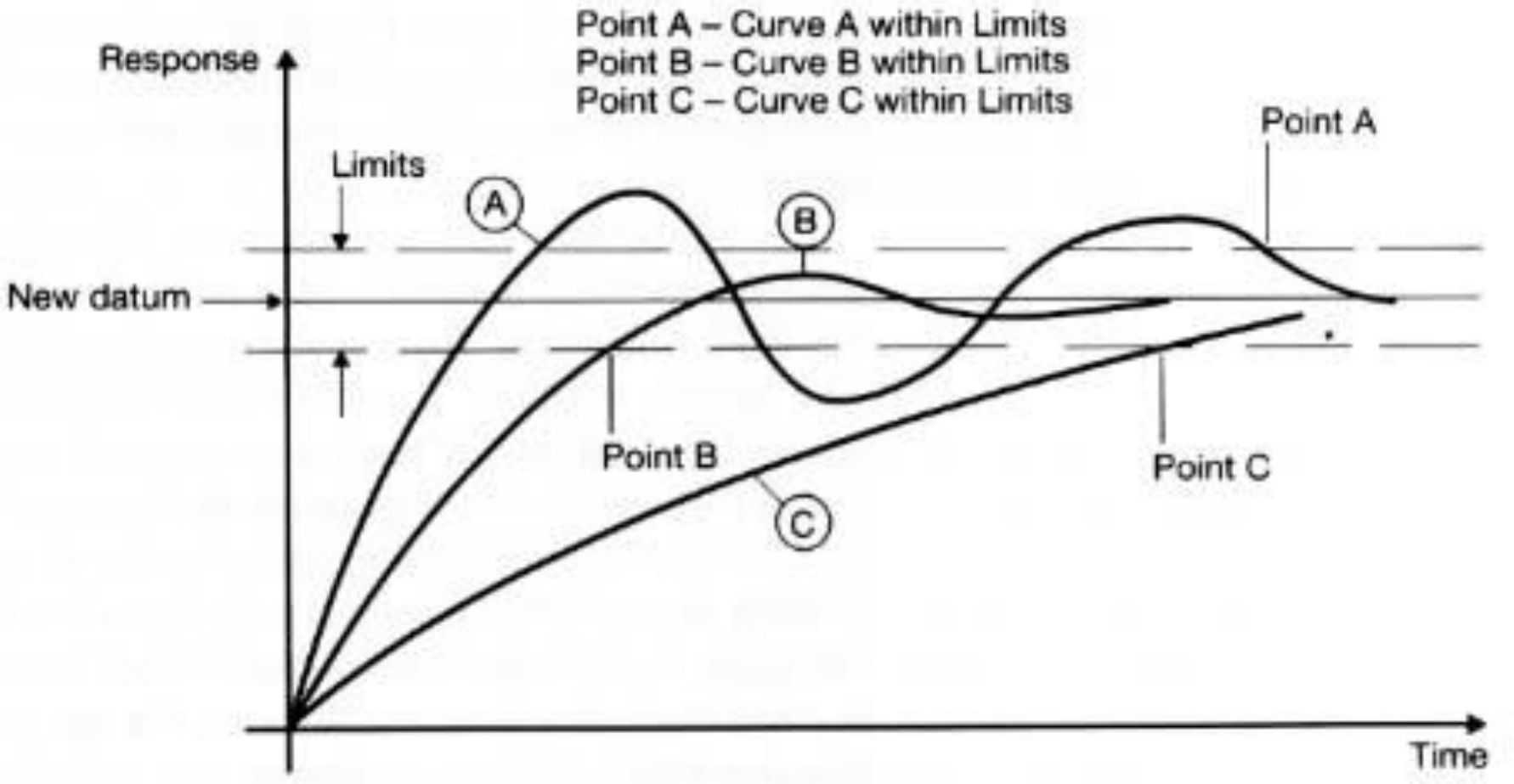


Fig. 3.12 Aerodynamically unstable aircraft.

Control for Stability Effect



Control System Characteristics



Control System Characteristics

- Controller characteristics is designed for stable flight performance
- A curve is normal characteristics, but causes overshoot and oscillation to convergence
- C curve responds slowly to convergence
- B curve is tuned to fast approach convergence in small overshoot